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(54) **FASTENER DELIVERY SYSTEM AND RELATED METHODS**

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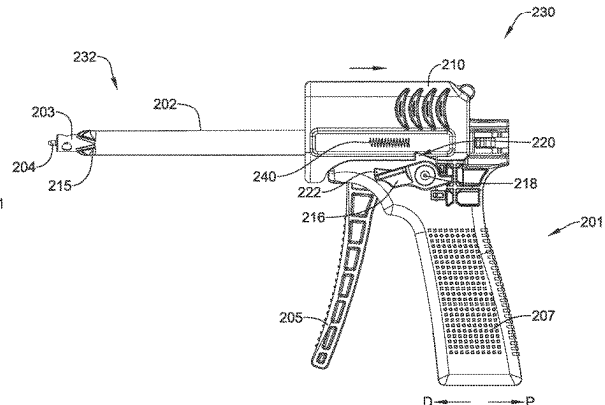
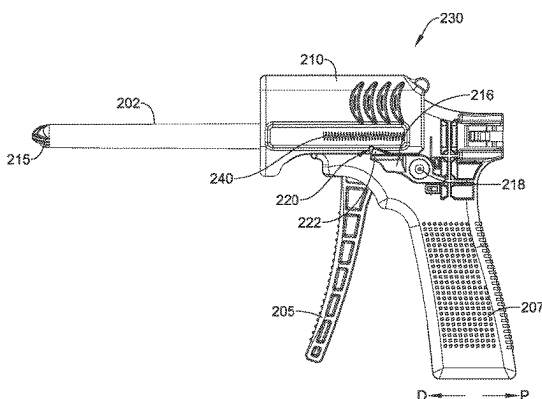
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(57) **ABSTRACT**

A fastener delivery tool is disclosed. An example fastener delivery tool comprises a sheath assembly including a sheath and a position retention member and a retraction assembly including a cover and a retractor member. The cover includes a lumen extending therein. The fastener delivery tool also includes and a handle assembly coupled to a proximal end of the sheath assembly, the handle assembly including a housing and an actuation member. Further, the sheath assembly extends through the lumen of the cover and the retractor member is configured to move relative to the handle between a first position in which the sheath assembly is disposed within a distal portion of the cover and a second position in which the sheath assembly extends out of the distal portion of the cover. Additionally, actuation of the actuation member moves the retractor member from the first position to the second position.

18 Claims, 21 Drawing Sheets



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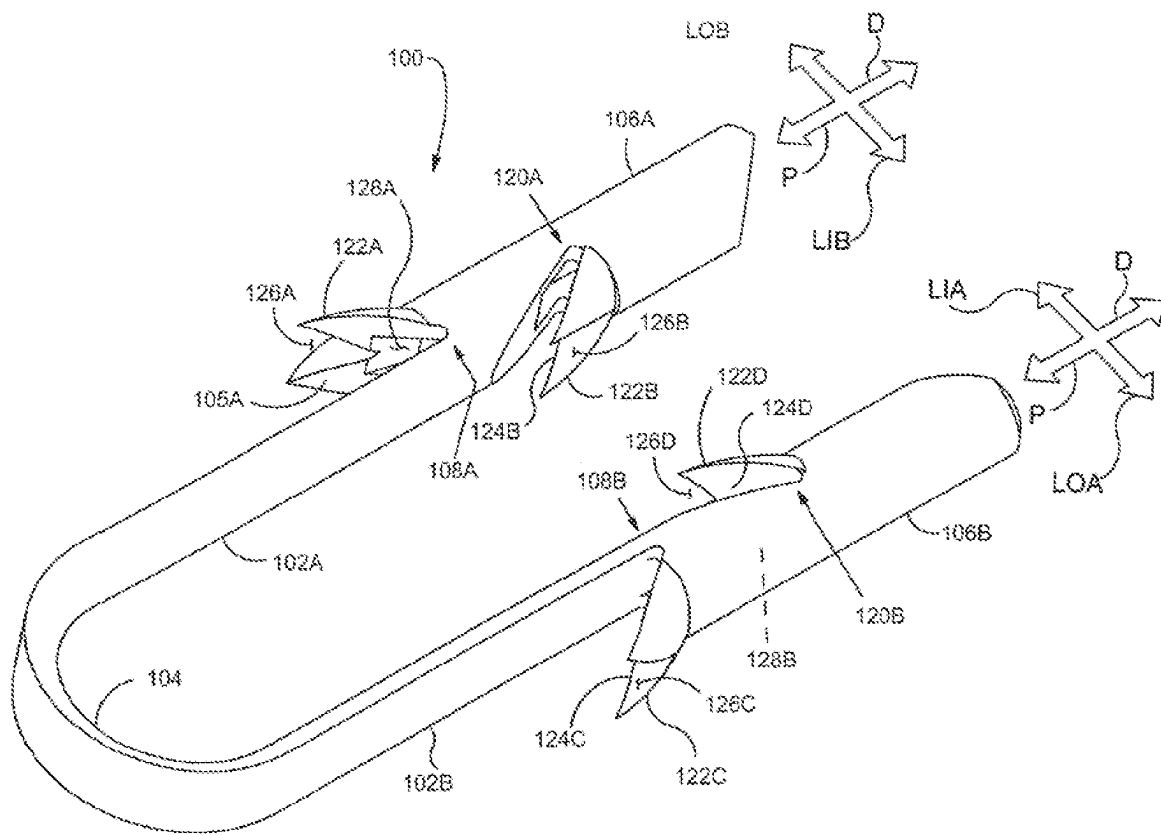


FIG. 1

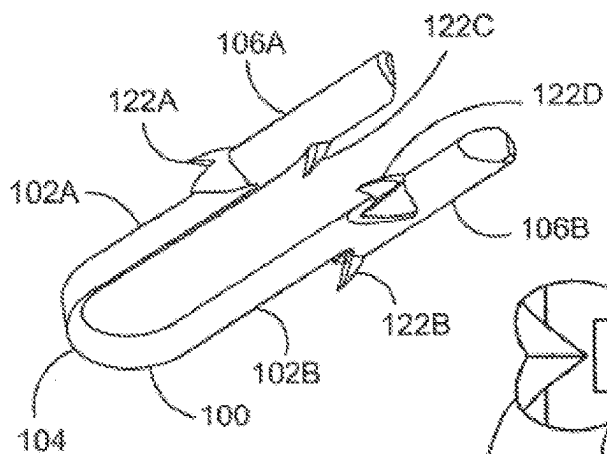


FIG. 2

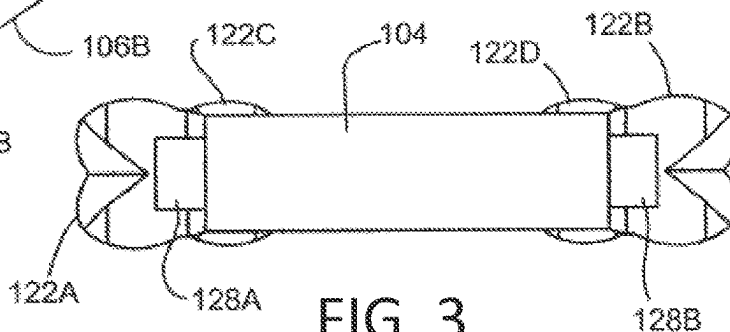


FIG. 3

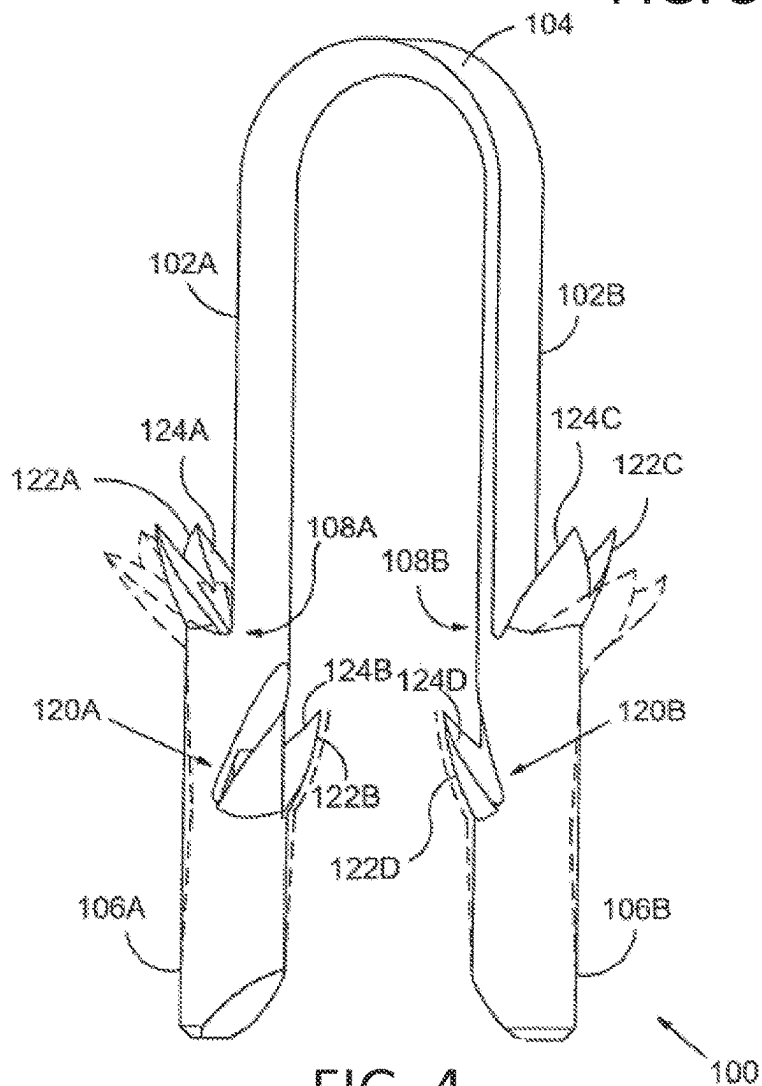


FIG. 4

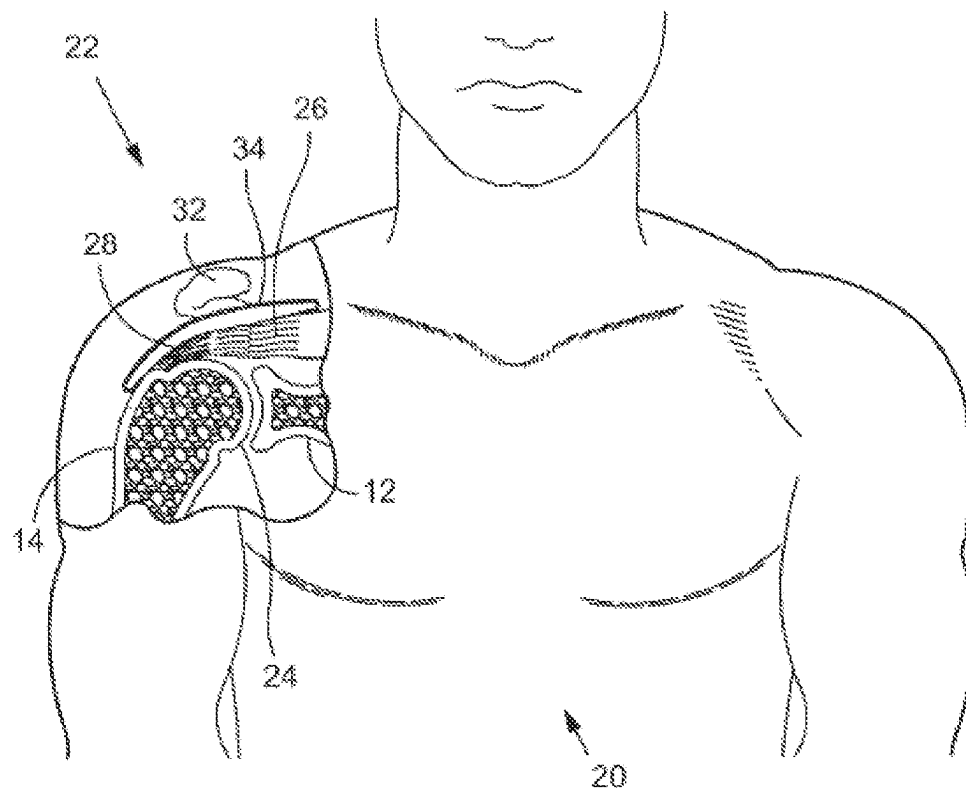


FIG. 5

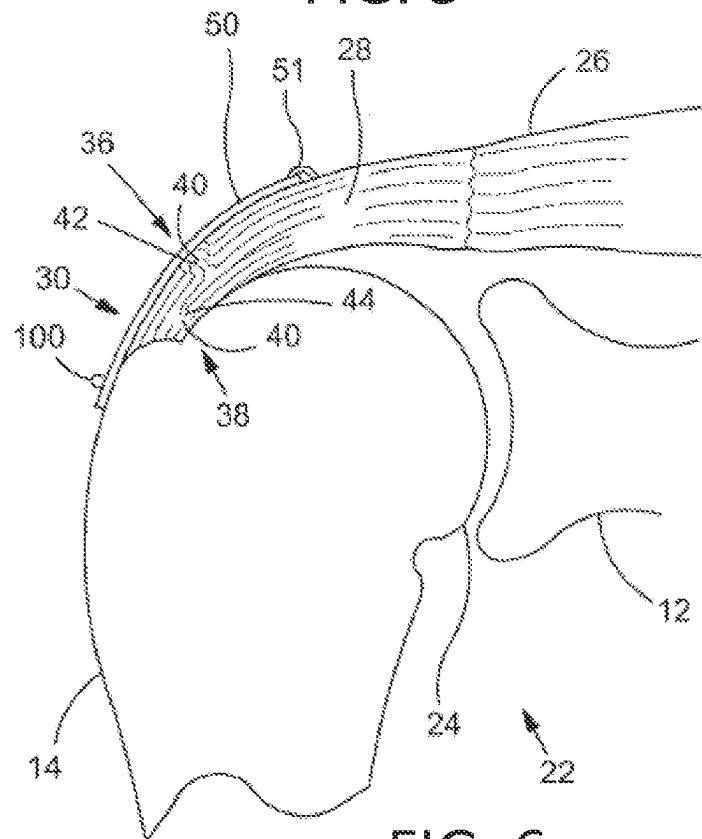


FIG. 6

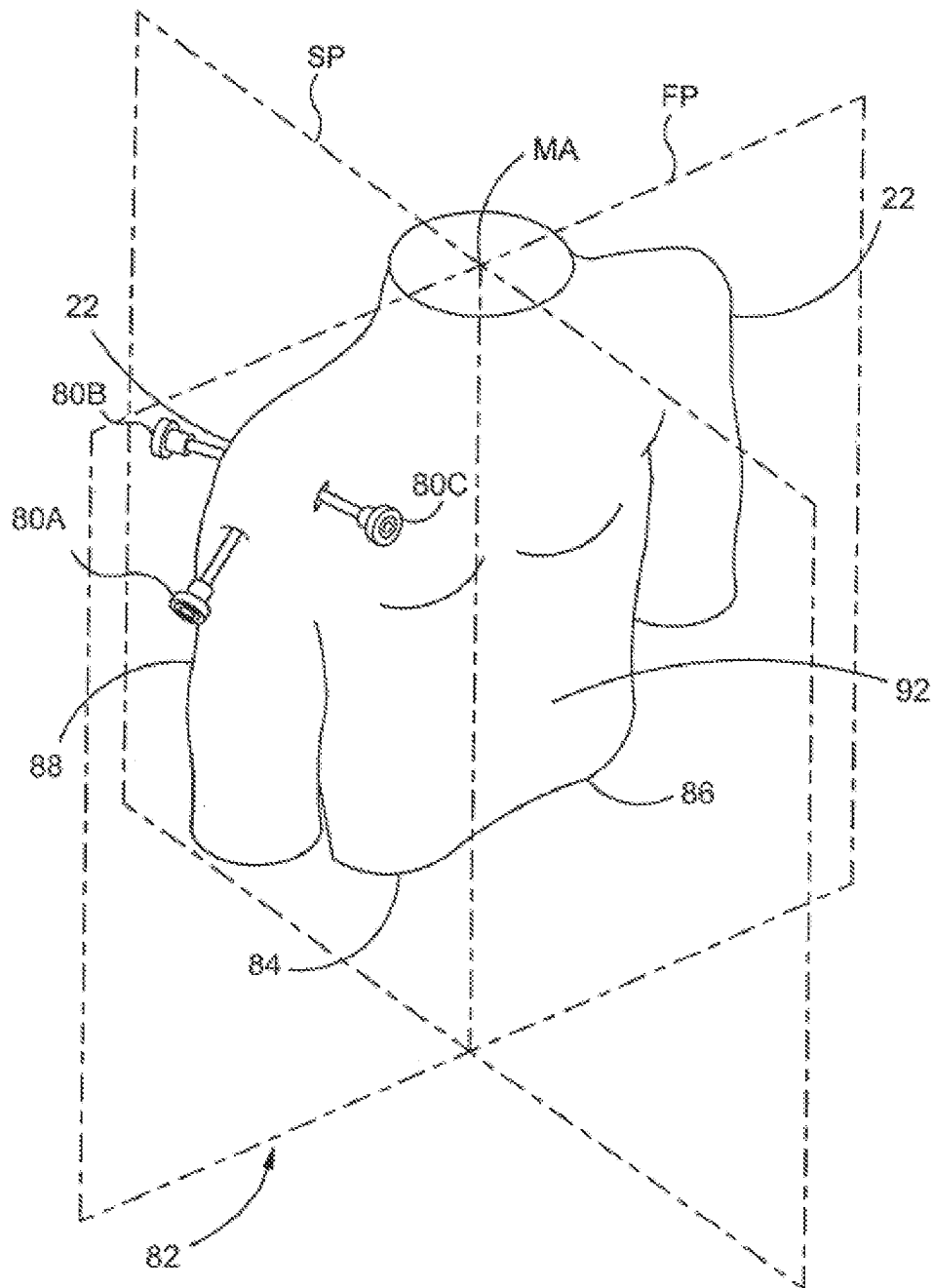


FIG. 7

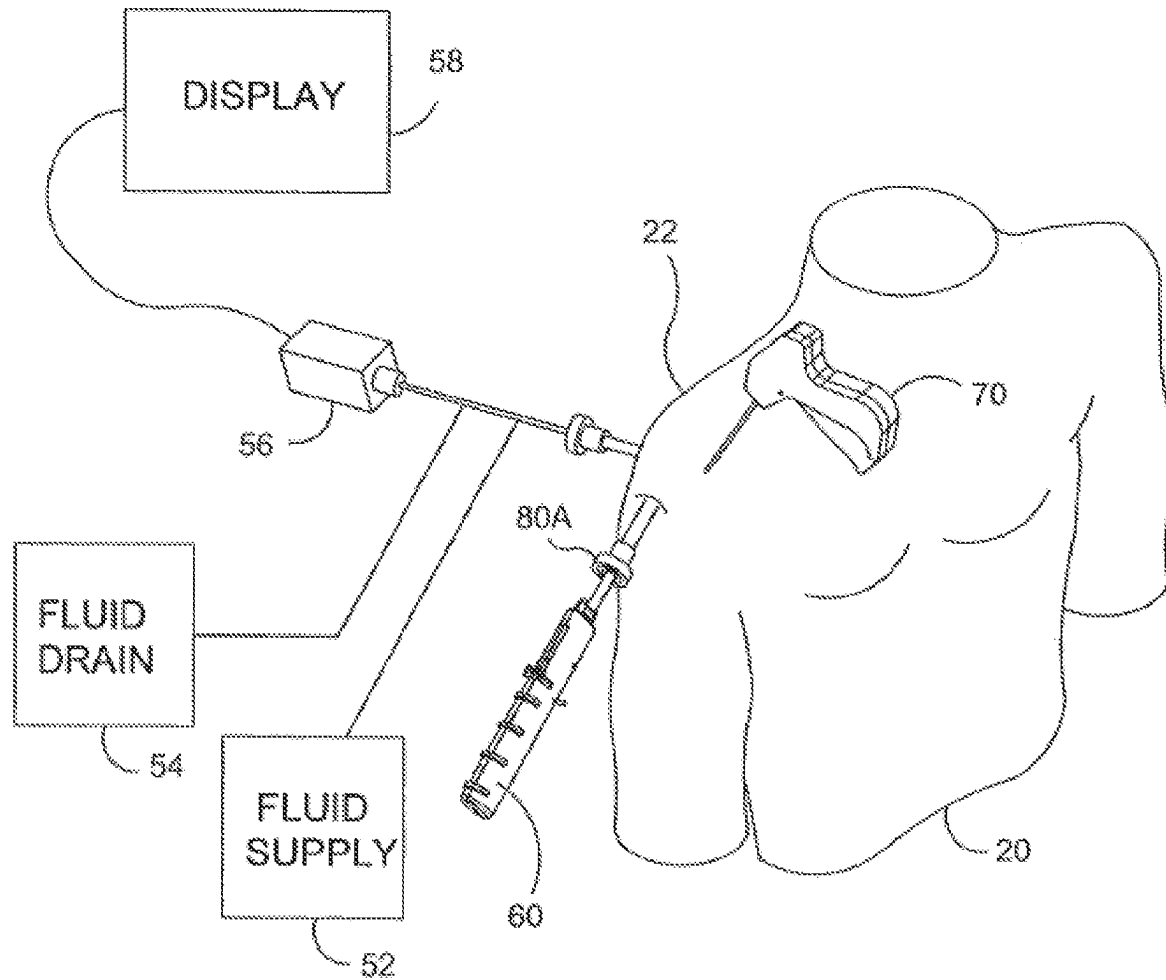


FIG. 8

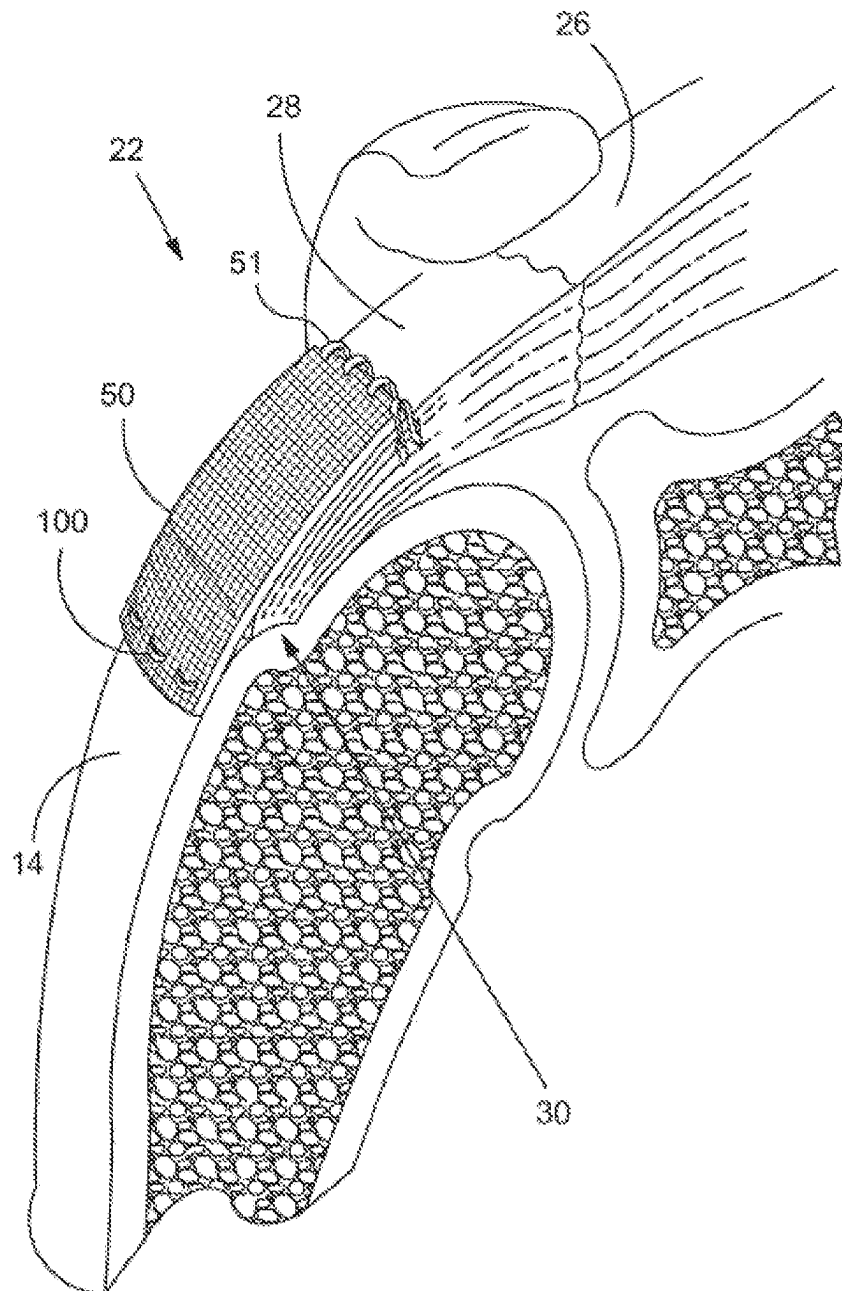


FIG. 9

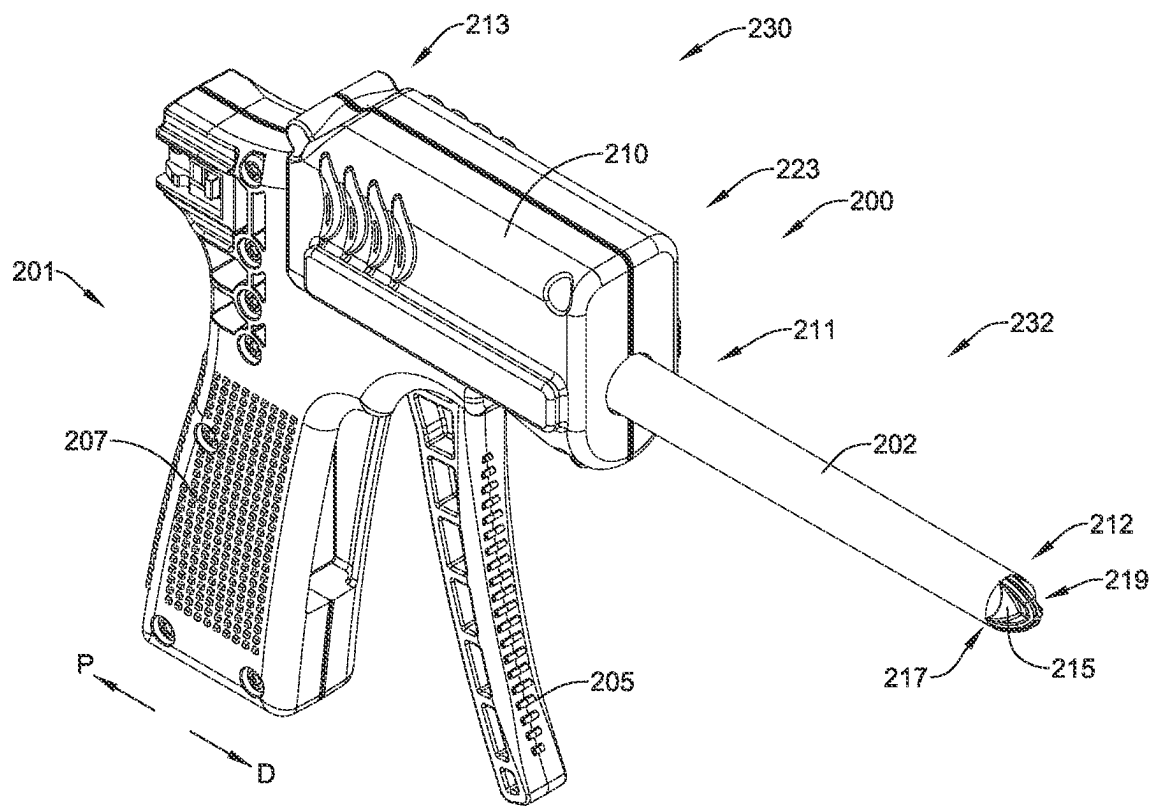


FIG. 10

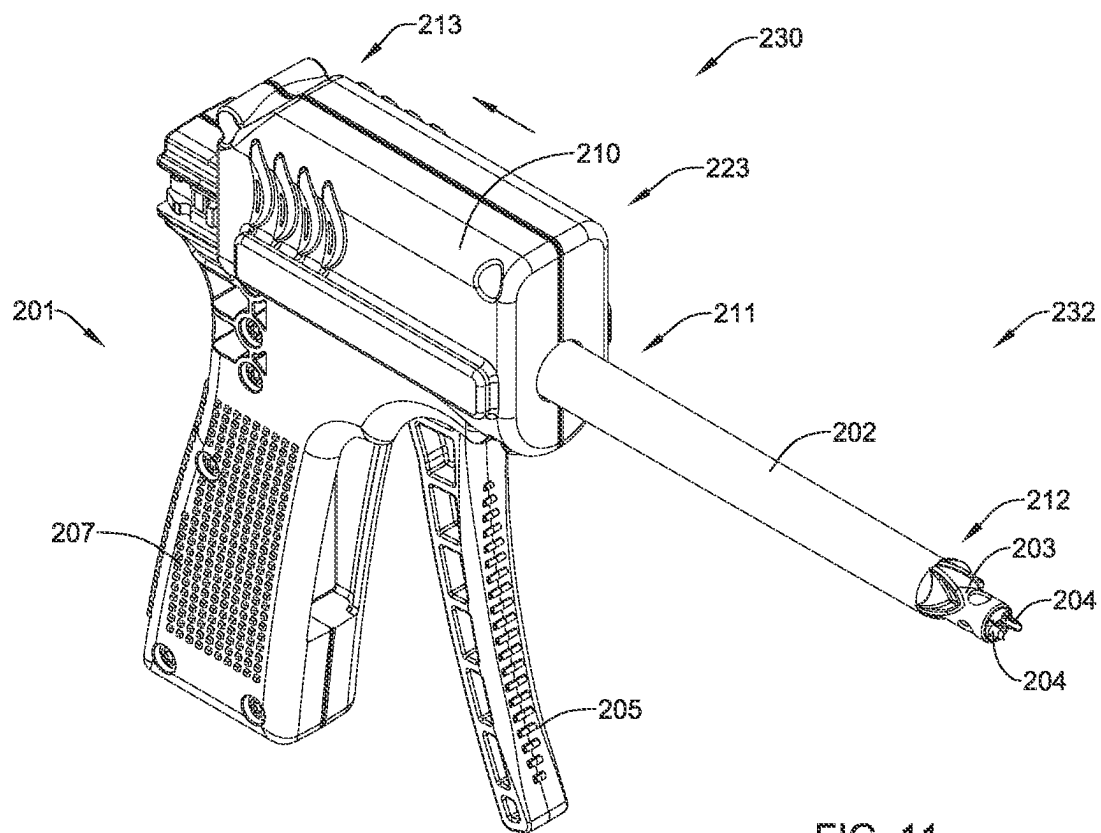


FIG. 11

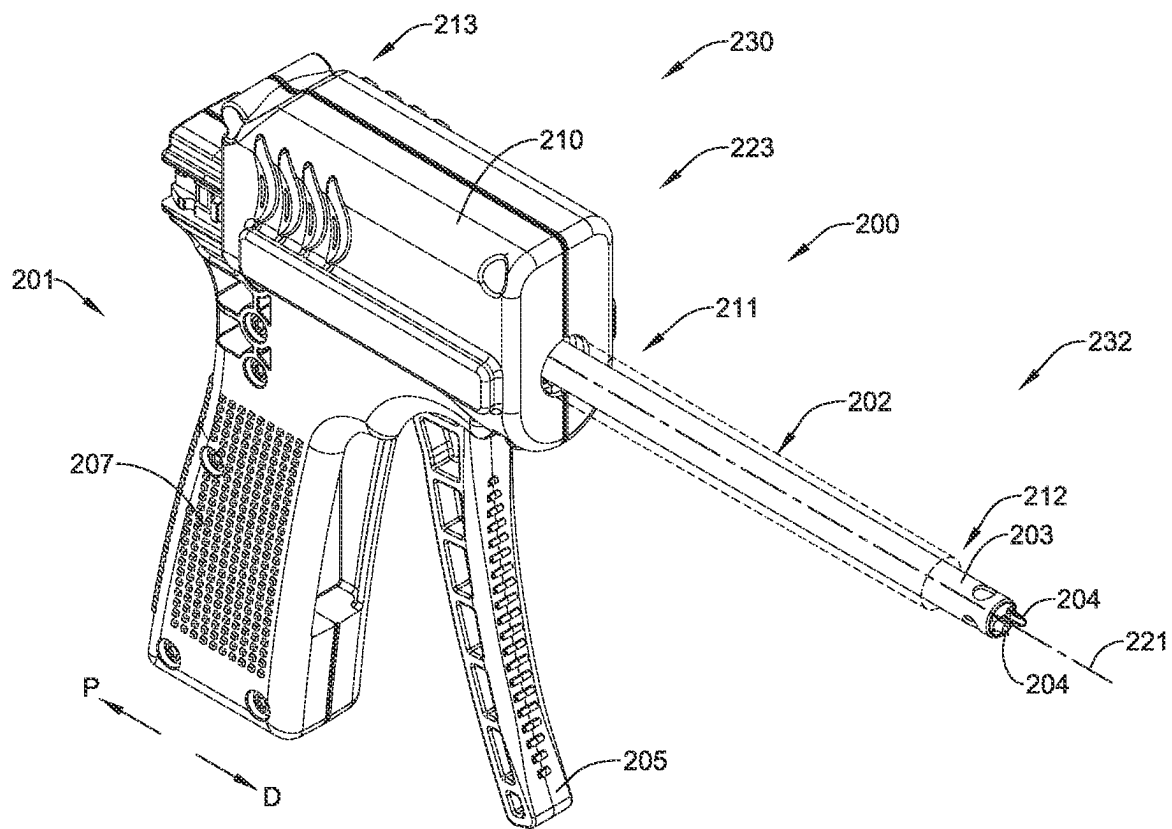


FIG. 12

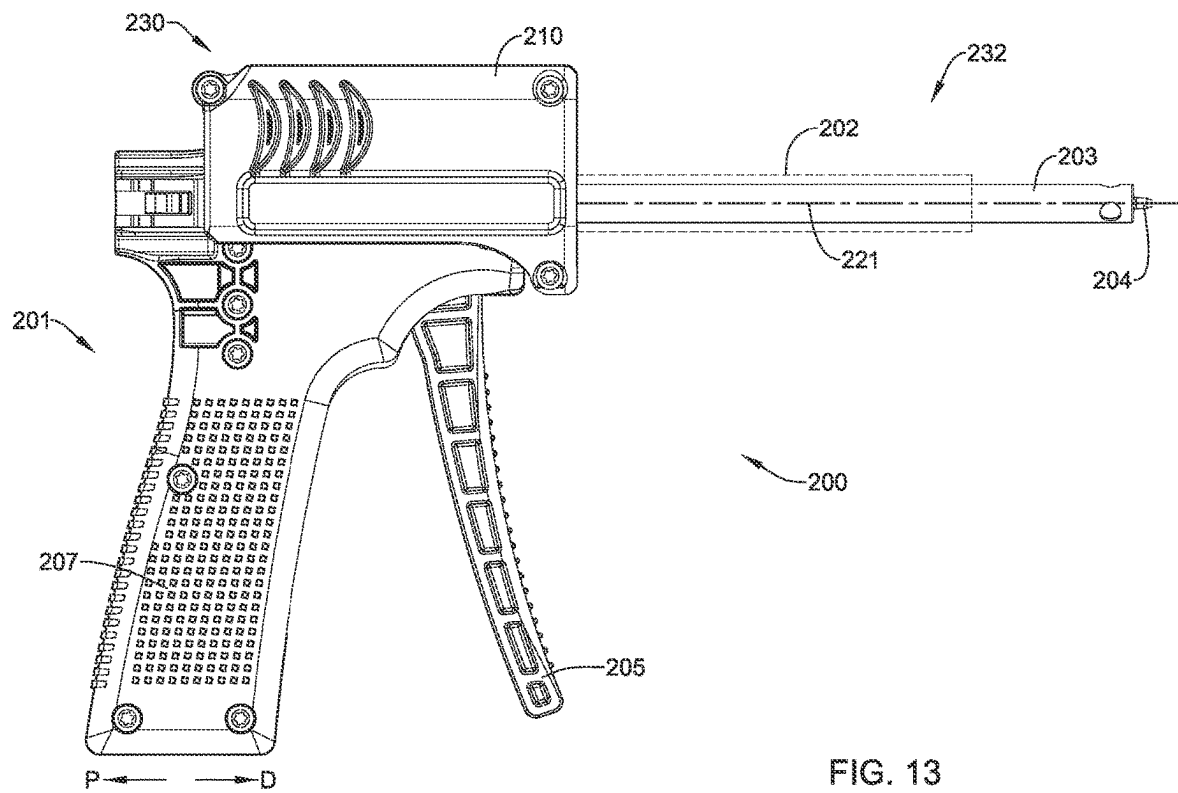


FIG. 13

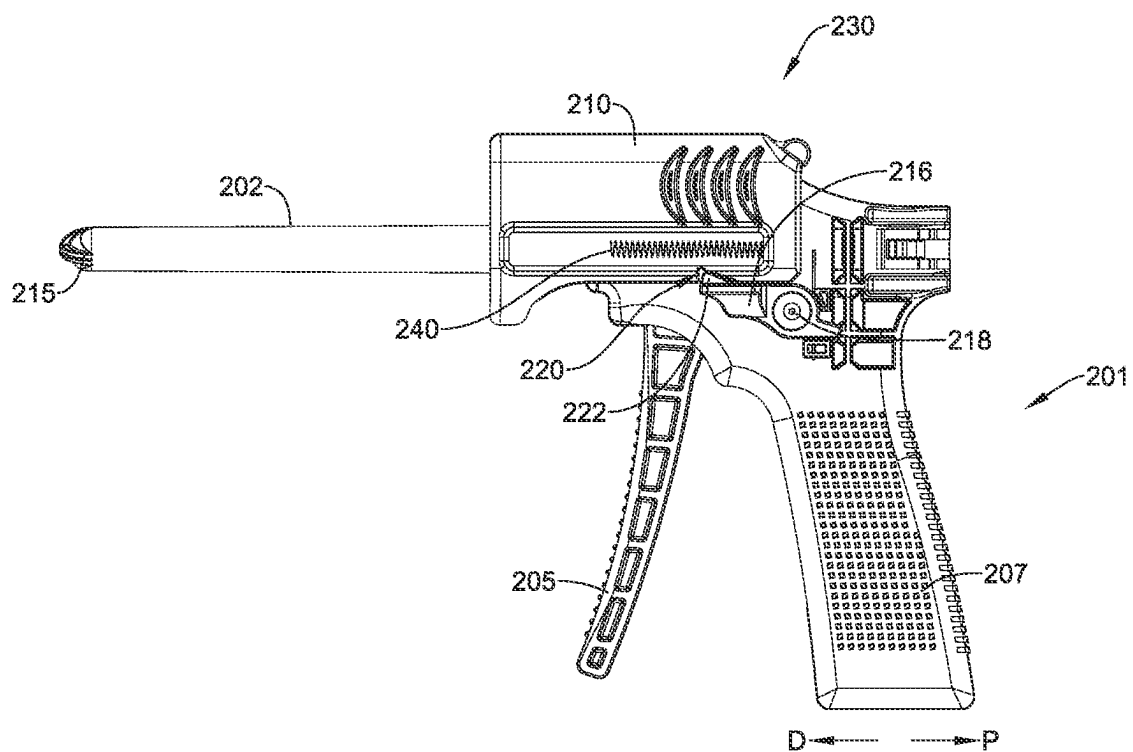


FIG. 14

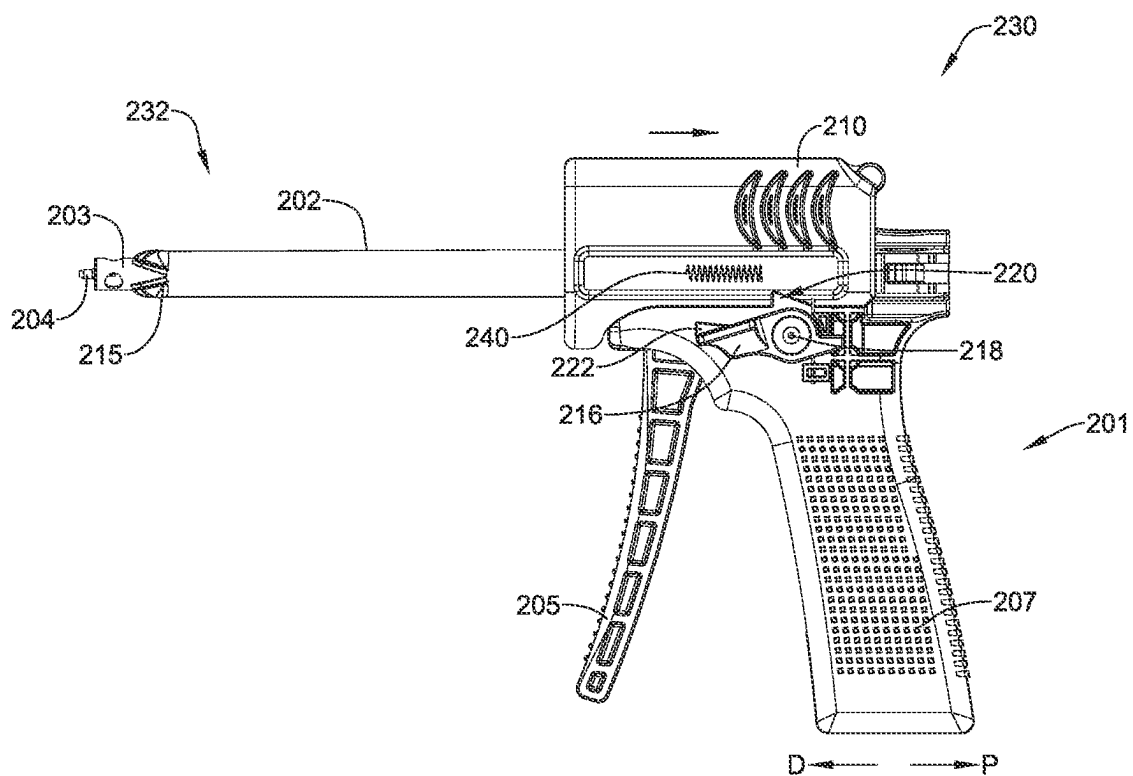


FIG. 15

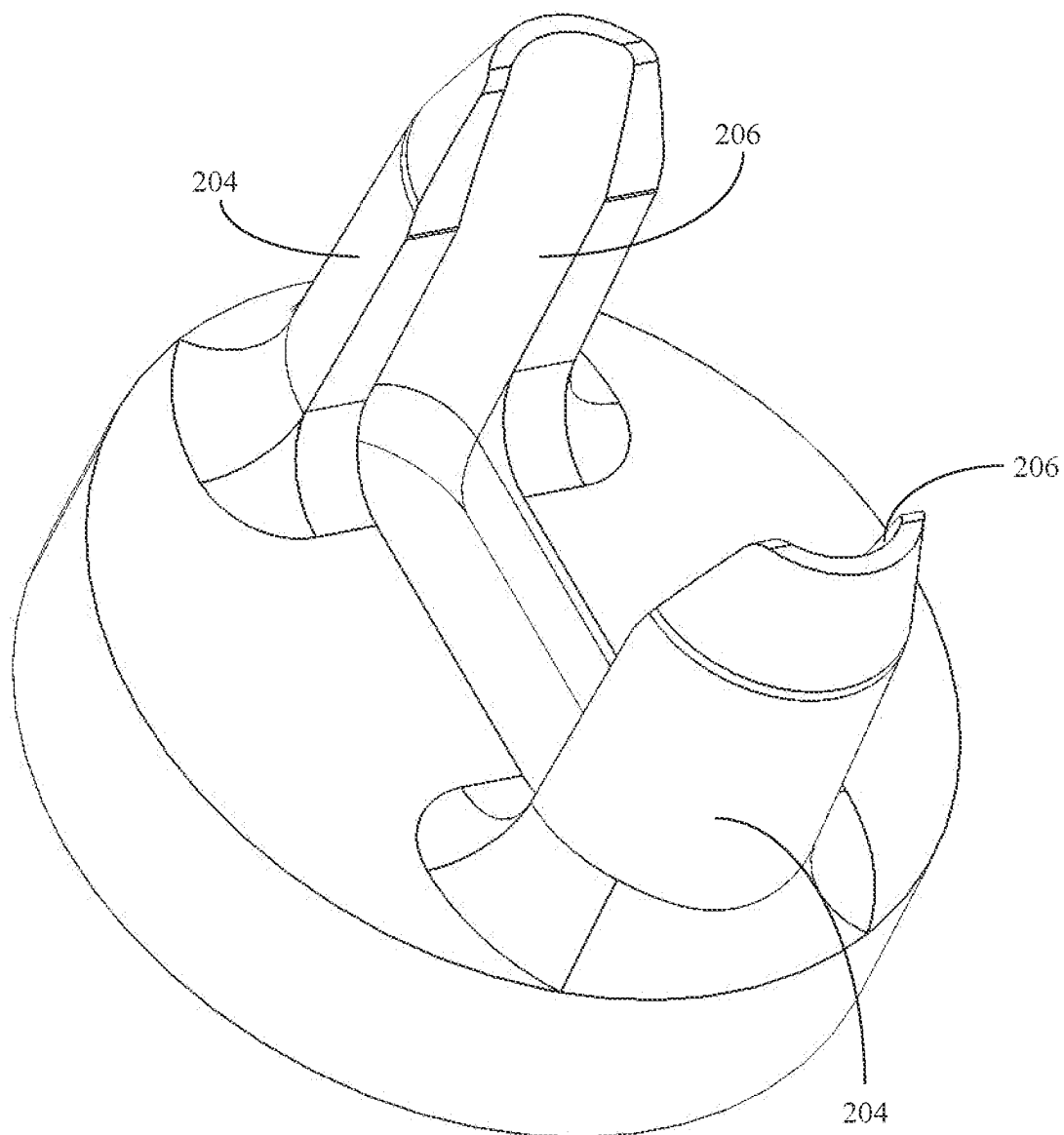
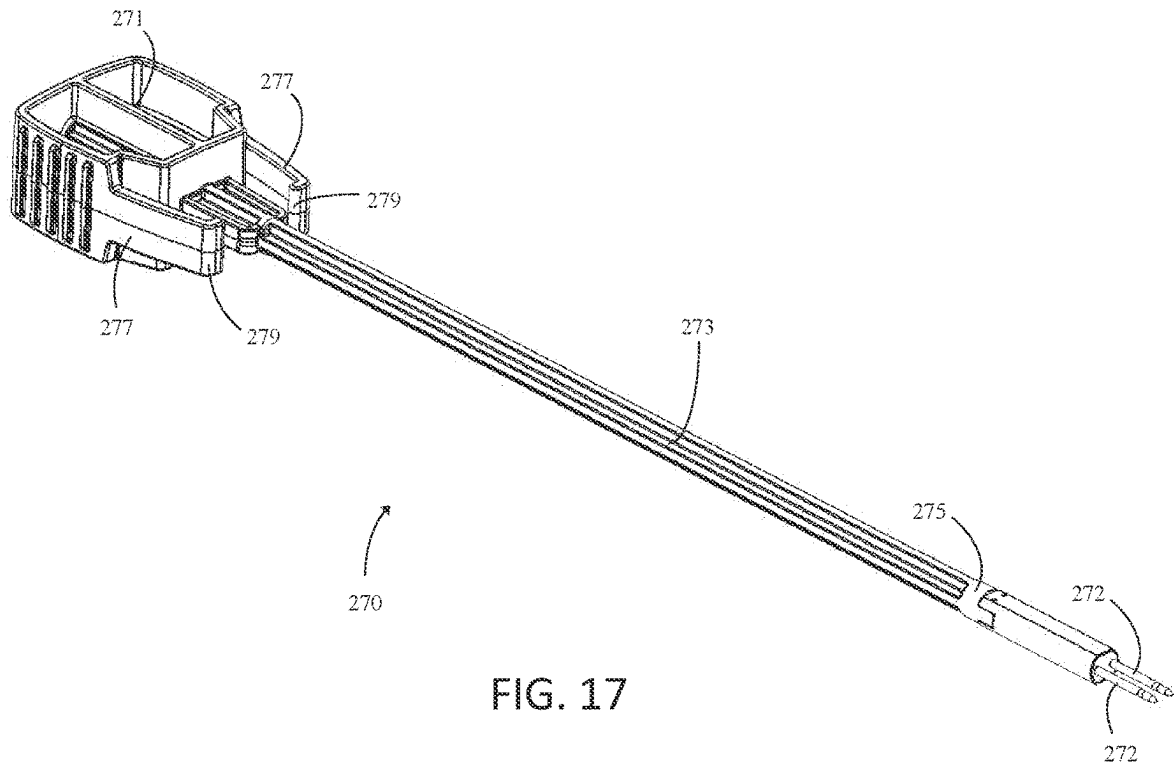


FIG. 16



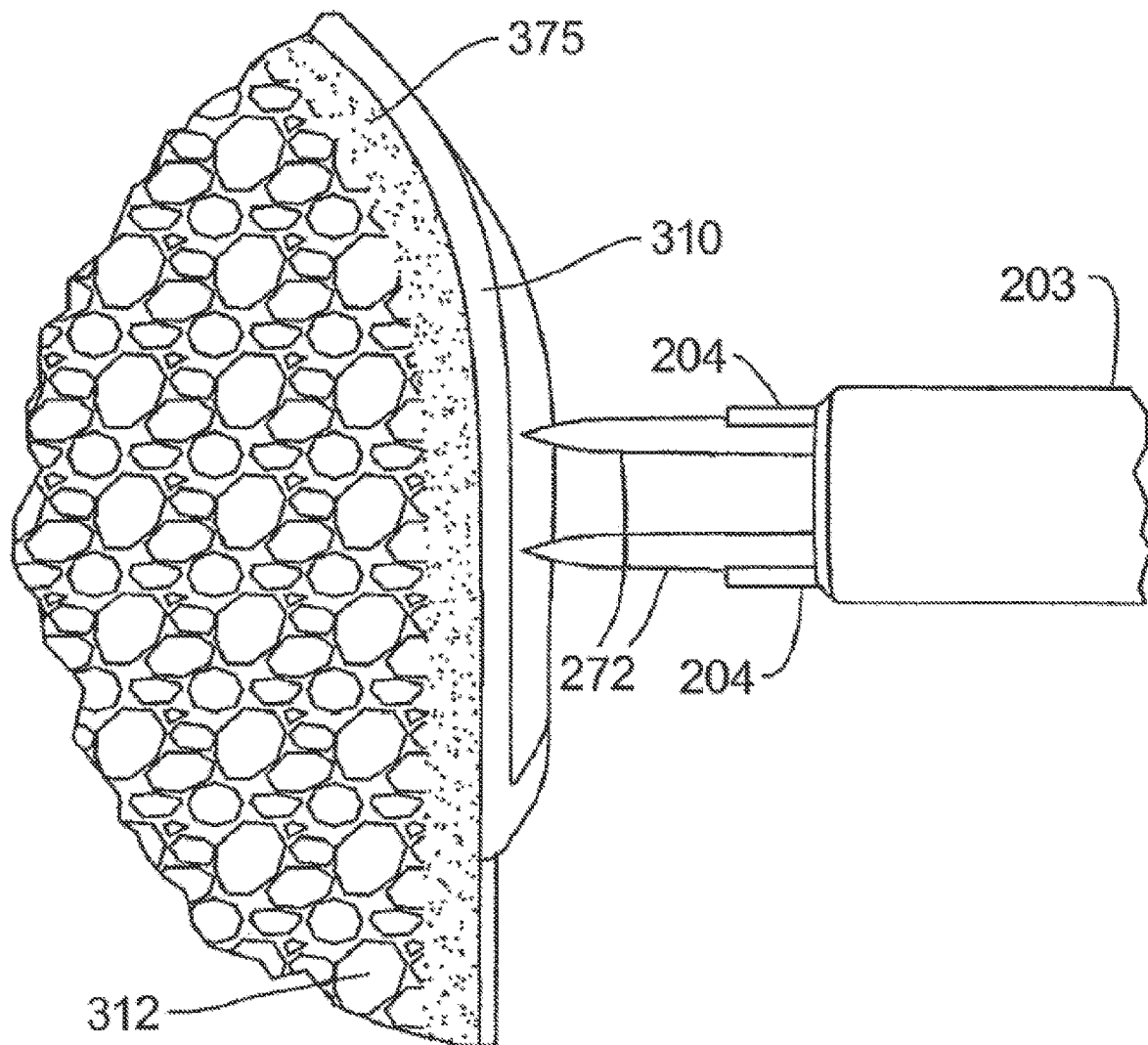


FIG. 18

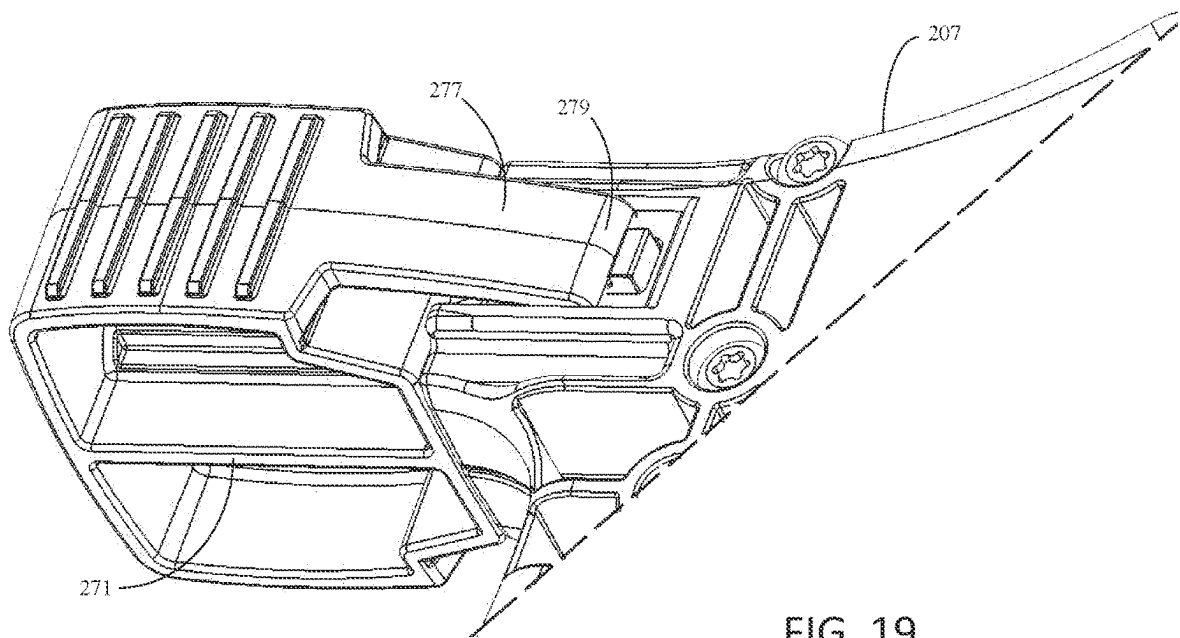


FIG. 19

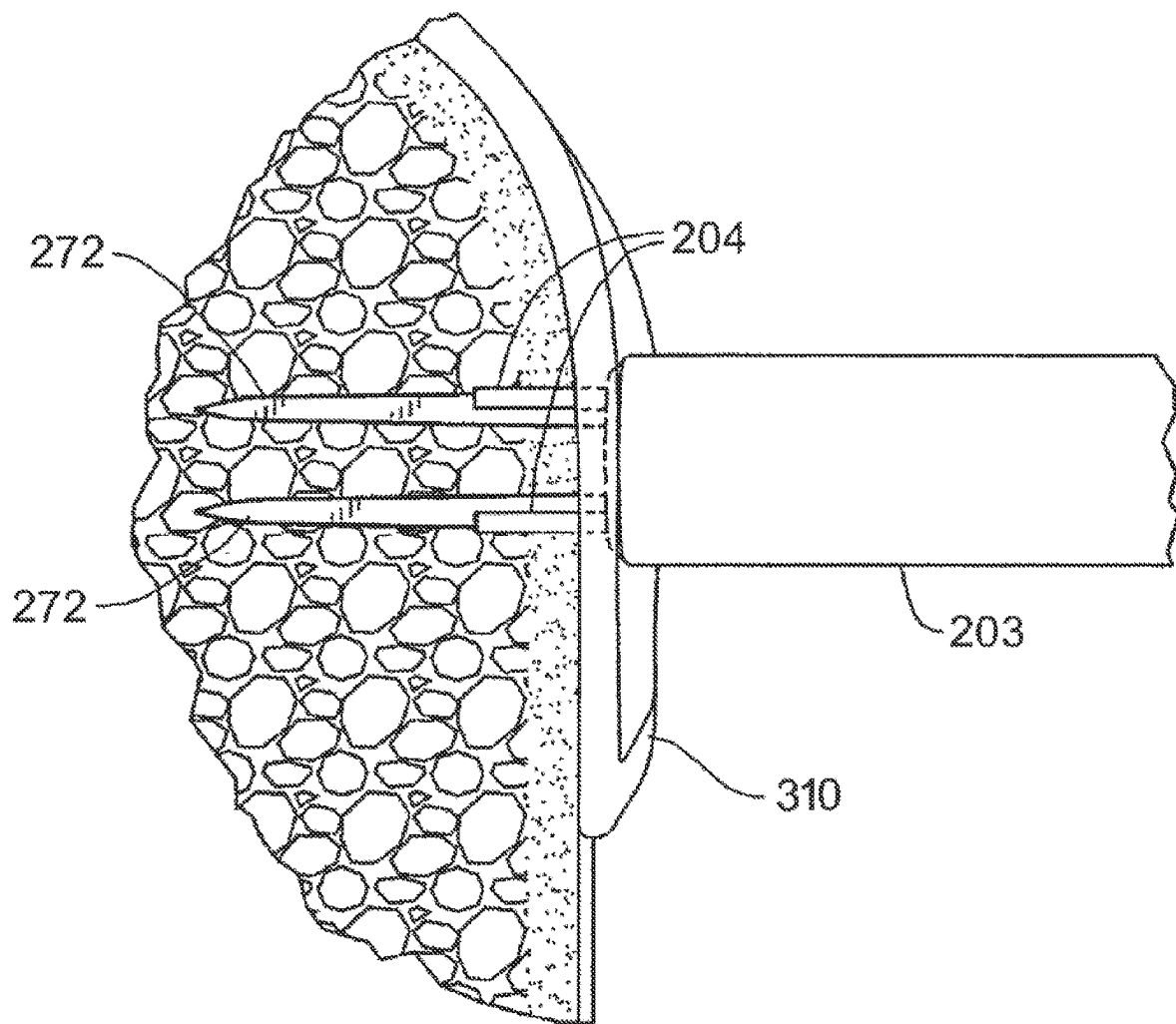


FIG. 20

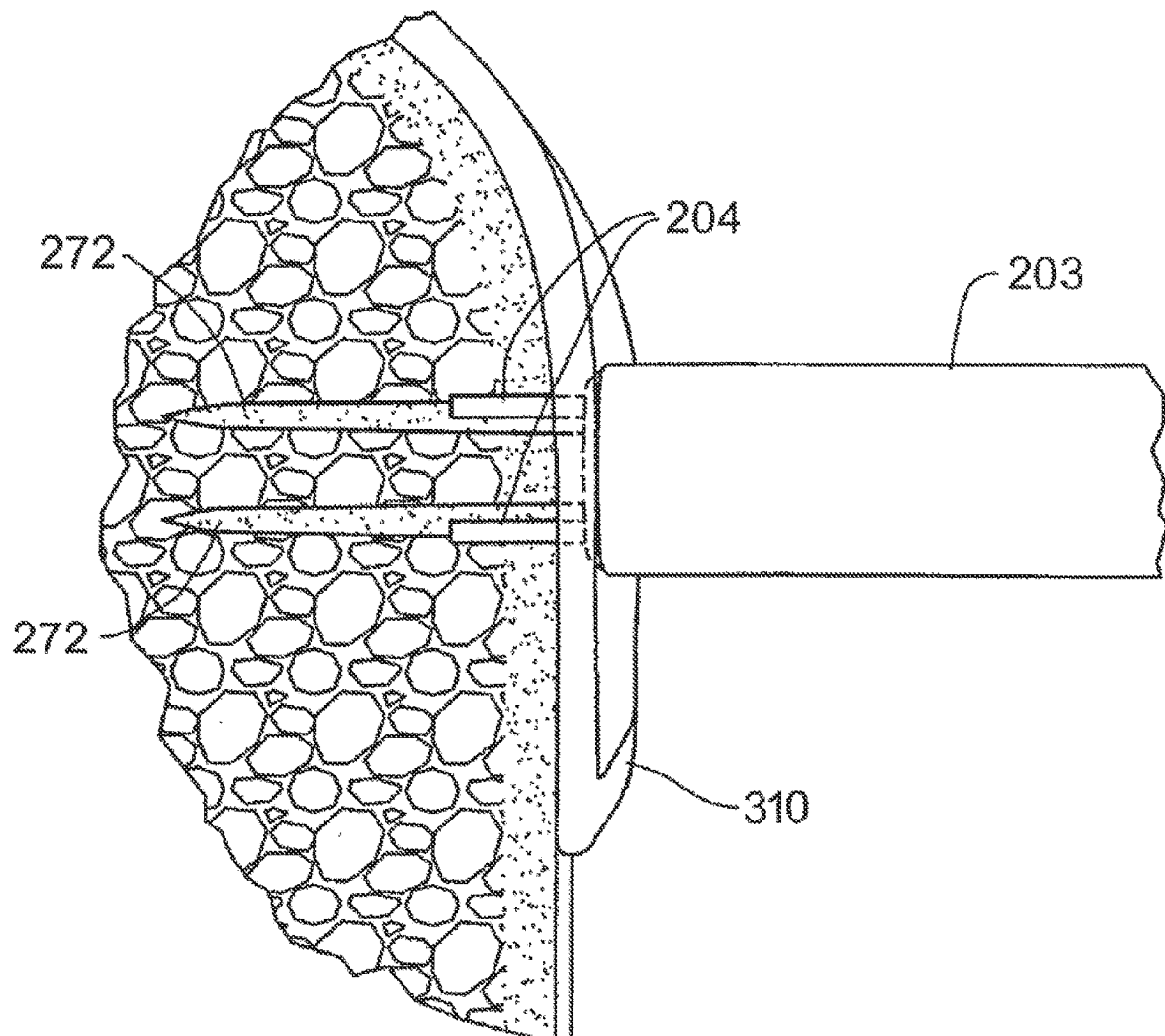
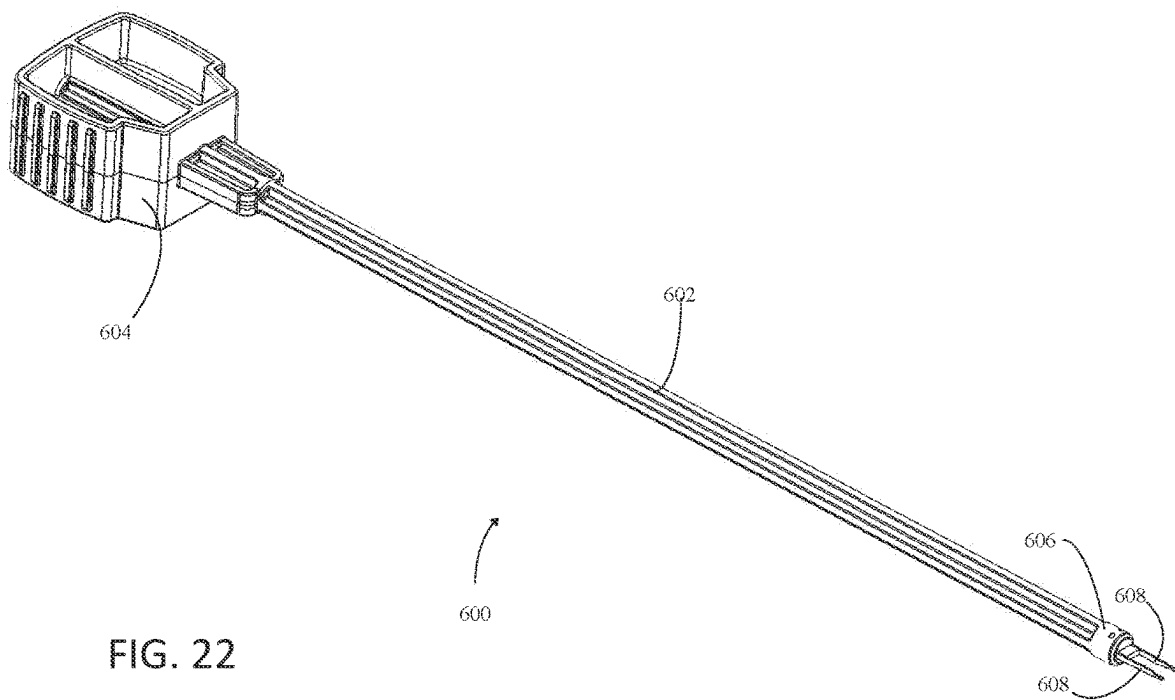


FIG. 21



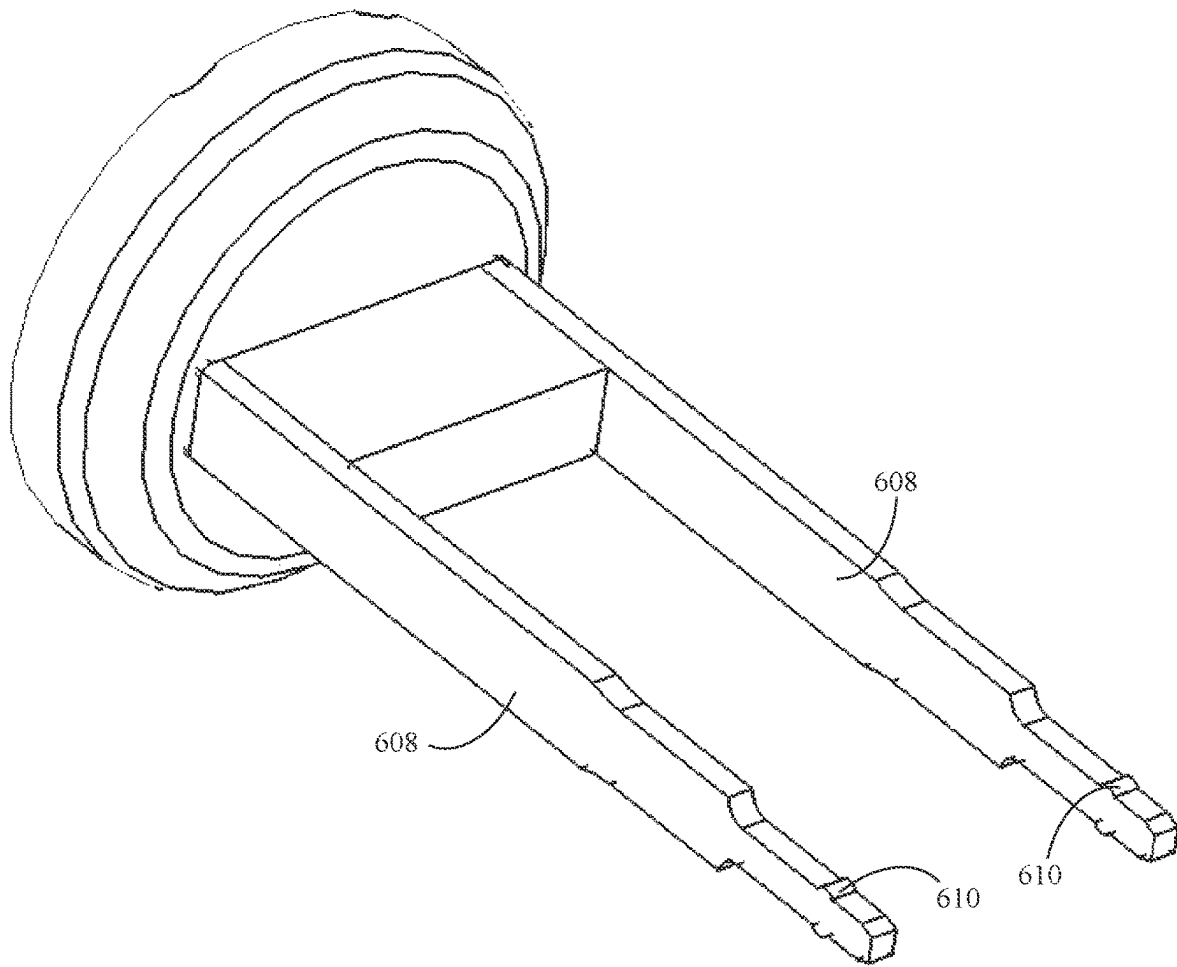


FIG. 23

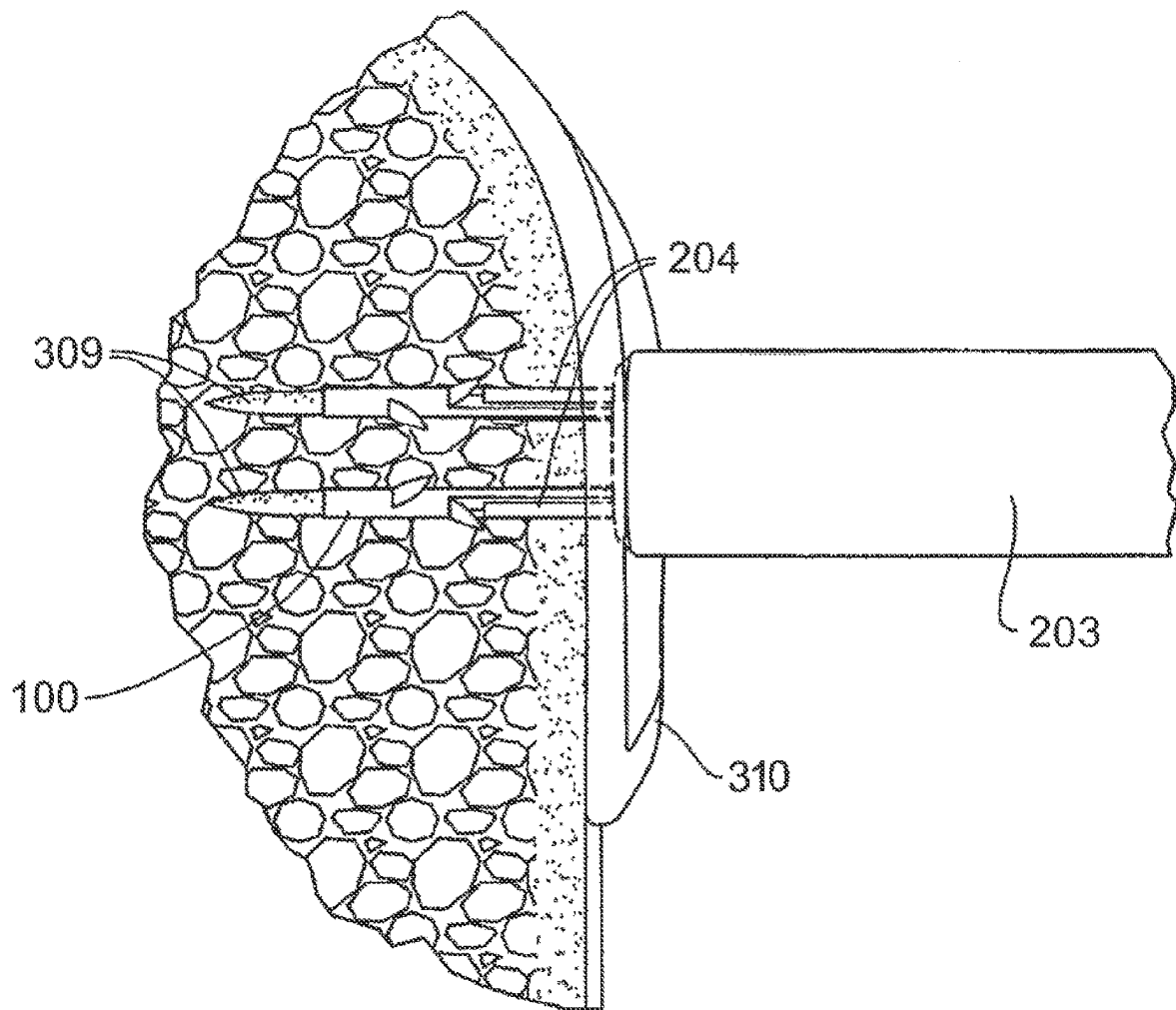


FIG. 24

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FASTENER DELIVERY SYSTEM AND RELATED METHODS

CROSS REFERENCE TO RELATED APPLICATIONS

This application is a continuation of U.S. patent application Ser. No. 16/934,977 filed Jul. 21, 2020, which is a continuation of U.S. patent application Ser. No. 15/394,350 filed Dec. 29, 2016, which claims the benefit of U.S. Provisional Patent Application Ser. No. 62/273,890 filed on Dec. 31, 2015, the disclosures of which are incorporated herein by reference. This application is also related to U.S. patent application Ser. No. 14/931,423 filed on Nov. 3, 2015, the disclosure of which is incorporated herein by reference.

TECHNICAL FIELD

The present disclosure pertains generally, but not by way of limitation, to medical devices, and methods for using medical devices. More particularly, the present disclosure pertains to devices for introducing and positioning implants within patients, and methods for using such devices.

BACKGROUND

With its complexity, range of motion and extensive use, a common soft tissue injury is damage to the rotator cuff or rotator cuff tendons. Damage to the rotator cuff is a potentially serious medical condition that may occur during hyperextension, from an acute traumatic tear or from overuse of the joint. Adequate procedures do not exist for repairing a partial thickness tear of less than 50% in the supraspinatus tendon. Current procedures attempt to alleviate impingement or make room for movement of the tendon to prevent further damage and relieve discomfort but do not repair or strengthen the tendon. Use of the still damaged tendon can lead to further damage or injury. There is an ongoing need to deliver and adequately position medical implants during an arthroscopic procedure in order to treat injuries to the rotator cuff, rotator cuff tendons, or other soft tissue or tendon injuries throughout a body.

SUMMARY OF THE DISCLOSURE

The disclosure describes various medical devices and methods for using medical devices to assist in delivering and positioning implants within a body. An example fastener delivery tool comprises a sheath assembly including a sheath and a position retention member and a retraction assembly including a cover and a retractor member. The cover includes a lumen extending therein. The fastener delivery tool also includes a handle assembly coupled to a proximal end of the sheath assembly, the handle assembly including a housing and an actuation member. Further, the sheath assembly extends through the lumen of the cover and the retractor member is configured to move relative to the handle between a first position in which the sheath assembly is disposed within a distal portion of the cover and a second position in which the sheath assembly extends out of the distal portion of the cover. Additionally, actuation of the actuation member moves the retractor member from the first position to the second position.

Alternatively or additionally, in another example, wherein the retraction assembly is biased to be in the second position.

Alternatively or additionally, in another example, further comprising a spring coupled to the retractor member.

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Alternatively or additionally, in another example, wherein the spring shifts the retractor member from the first position to the second position when the actuation member is actuated.

5 Alternatively or additionally, in another example, wherein the cover includes one or more leaflets positioned on a distal portion thereof.

Alternatively or additionally, in another example, wherein at least one of the one or more leaflets includes a proximal portion and a distal portion, and wherein at least one of the one or more leaflets are tapered from the proximal portion to the distal portion.

Alternatively or additionally, in another example, wherein the leaflets are biased in a closed position.

15 Alternatively or additionally, in another example, wherein the leaflets are configured to expand radially outward as the sheath assembly slides from the first position to the second position.

Alternatively or additionally, in another example, wherein shifting the retractor member between the first position and the second position uncovers the sheath assembly in vivo.

Alternatively or additionally, in another example, wherein a proximal portion of the cover is attached to the retractor member.

25 Alternatively or additionally, in another example, wherein the cover is coaxial with the sheath assembly.

Alternatively or additionally, in another example, wherein the actuation member includes a projection configured to engage with a recess disposed along the retractor member.

30 Alternatively or additionally, in another example, wherein the actuation member is designed to rotate relative to the housing.

Alternatively or additionally, in another example, wherein actuation of the actuation member rotates the actuation member about an attachment point disposed along the housing, and wherein rotation of the actuation member about the attachment point is designed to disengage the projection from the recess.

40 Alternatively or additionally, in another example, wherein a longitudinal axis of the handle housing is aligned with a longitudinal axis of the sheath, and wherein the retractor member shifts along both the longitudinal axis of the sheath and the longitudinal axis of the housing.

Another example fastener delivery tool comprises a handle assembly including a housing, a retractor member and an actuation member; and a sheath assembly including a sheath, a cover and a position retention member. Additionally, the sheath extends within a lumen of the cover, a proximal portion of the sheath assembly is coupled to the handle assembly, the retractor member is configured to slide along the handle between a first position in which the sheath assembly is disposed within a distal portion of the cover and a second position in which the sheath assembly extends out of the distal portion of the cover and actuation of the actuation member slides the retractor member from the first position to the second position.

Alternatively or additionally, in another example, wherein the cover includes one or more leaflets positioned on a distal portion thereof.

60 Alternatively or additionally, in another example, wherein the retractor member is biased to be in the second position.

Alternatively or additionally, in another example, wherein shifting the retractor member between the first position and the second position extends the sheath out of the distal portion of the cover in vivo.

An example method for deploying a fastener comprises positioning a fastener delivery tool adjacent a target site. The

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fastener delivery tool comprises a sheath assembly including a sheath and a position retention member; a retraction assembly including a cover and a retractor member, wherein the cover includes a lumen extending therein; and a handle assembly coupled to a proximal end of the sheath assembly, the handle assembly including a housing, a trigger and an actuation member. Further, the sheath assembly extends through the lumen of the cover and the retractor member is configured to move along the handle between a first position in which the sheath assembly is disposed within a distal portion of the cover and a second position in which the sheath assembly extends out of the distal portion of the cover. Additionally, actuation of the actuation member moves the retractor member from the first position to the second position. The method also includes deploying a staple along a target site.

The above summary of some examples is not intended to describe each disclosed example device, component, or method or every implementation of the present disclosure. The Brief Description of the Drawings, and Detailed Description, which follow, more particularly exemplify these examples, but are also intended as exemplary and not limiting.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 is a perspective view illustrating an exemplary tissue fastener or staple in accordance with the present disclosure;

FIG. 2 is a an alternative perspective view of the tissue fastener or staple of FIG. 1 illustrating other features in accordance with the present disclosure;

FIG. 3 is a top plan view of the tissue fastener or staple of FIG. 1 illustrating the laterally extending legs having lumens for receiving the stakes of a delivery device of the present disclosure;

FIG. 4 is a front plan view of the tissue fastener or staple of FIG. 1 illustrating in phantom flexing of the barbs and legs of the staple in response to grasping of tissue in one example of the present disclosure;

FIG. 5 is a stylized anterior view of a shoulder including a humerus and a scapula;

FIG. 6 is a stylized anterior view of a shoulder depicting the head of the humerus shown mating with the glenoid fossa of the scapula at a glenohumeral joint and a sheet-like material is affixed to the tendon;

FIG. 7 is a stylized perspective view showing a portion of the body of a human patient divided into quadrants by planes for descriptive purposes herein;

FIG. 8 is a stylized perspective view illustrating an exemplary procedure for arthroscopic treatment of a shoulder of a patient in accordance with one example of the present disclosure;

FIG. 9 is a stylized perspective view of a shoulder including a supraspinatus having a distal tendon with a sheet-like material affixed thereto;

FIG. 10 is a perspective view illustrating an example fastener delivery device in accordance with one example of the present disclosure;

FIG. 11 is a perspective view illustrating an example fastener delivery device in accordance with one example of the present disclosure;

FIG. 12 is a perspective view illustrating an example fastener delivery device in accordance with one example of the present disclosure;

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FIG. 13 is a plan view illustrating an example fastener delivery device in accordance with one example of the present disclosure;

FIG. 14 is a plan view illustrating an example fastener delivery device in accordance with one example of the present disclosure;

FIG. 15 is a plan view illustrating an example fastener delivery device in accordance with one example of the present disclosure;

FIG. 16 is a schematic illustration depicting position retention members of an example fastener delivery device in accordance with one example of the present disclosure;

FIG. 17 is a schematic illustration depicting an example pilot hole forming insert in accordance with one example of the present disclosure;

FIG. 18 is a cross section view depicting an example fastener delivery device in accordance with one example of the present disclosure positioned at an implant site;

FIG. 19 is a schematic illustration depicting a proximal portion of an example fastener delivery device in accordance with one example of the present disclosure and a proximal head of an example insert when the insert is received within a sheath of the example fastener delivery device and after force has been applied to the proximal head to move the example insert in the distal direction;

FIG. 20 is a cross section view depicting an example fastener delivery device in accordance with one example of the present disclosure positioned at an implant site after pilot hole forming members have been driven into tissue at the implant site;

FIG. 21 is a cross section view depicting an example fastener delivery device in accordance with one example of the present disclosure positioned at an implant site after pilot hole forming members have been driven into and removed from tissue at the implant site;

FIG. 22 is a schematic illustration depicting an example staple delivery insert in accordance with one example of the present disclosure;

FIG. 23 is a schematic illustration of a distal portion of an example staple delivery insert in accordance with one example of the present disclosure; and

FIG. 24 is a cross section view depicting an example fastener delivery device in accordance with one example of the present disclosure positioned at an implant site after an example fastener has been deployed into pilot holes formed in tissue at the implant site.

DETAILED DESCRIPTION

The following description should be read with reference to the drawings, which are not necessarily to scale, wherein like reference numerals indicate like elements throughout the several views. The detailed description and drawings are intended to illustrate but not limit the claimed invention. Those skilled in the art will recognize that the various elements described and/or shown may be arranged in various combinations and configurations without departing from the scope of the disclosure. The detailed description and drawings illustrate examples of the claimed invention.

Definitions of certain terms are provided below and shall be applied, unless a different definition is given in the claims or elsewhere in this specification.

All numeric values are herein assumed to be modified by the term “about,” whether or not explicitly indicated. The term “about” generally refers to a range of numbers that one of skill in the art would consider equivalent to the recited value (i.e., having the same or substantially the same func-

tion or result). In many instances, the terms “about” may include numbers that are rounded to the nearest significant figure. Other uses of the term “about” (i.e., in a context other than numeric values) may be assumed to have their ordinary and customary definition(s), as understood from and consistent with the context of the specification, unless otherwise specified.

The recitation of numerical ranges by endpoints includes all numbers within that range (e.g., 1 to 5 includes 1, 1.5, 2, 2.75, 3, 3.80, 4, and 5).

As used in this specification and the appended claims, the singular forms “a,” “an,” and “the” include or otherwise refer to singular as well as plural referents, unless the content clearly dictates otherwise. As used in this specification and the appended claims, the term “or” is generally employed to include “and/or,” unless the content clearly dictates otherwise.

It is noted that references in the specification to “an embodiment,” “some embodiments,” “other embodiments,” “an example,” “some examples,” “other examples,” etc., indicate that the embodiment(s) and/or example(s) described may include a particular feature, structure, or characteristic, but every embodiment may not necessarily include the particular feature, structure, or characteristic. Moreover, such phrases are not necessarily referring to the same embodiment and/or example. Further, when a particular feature, structure, or characteristic is described in connection with an embodiment and/or example, it would be within the knowledge of one skilled in the art to affect such feature, structure, or characteristic in connection with other embodiments and/or examples, whether or not explicitly described, unless clearly stated to the contrary. That is, the various individual features described below, even if not explicitly shown in a particular combination, are nevertheless contemplated as being combinable or able to be arranged with each other to form other additional embodiments and/or examples or to complement and/or enrich the described embodiment(s) and/or example(s), as would be understood by one of ordinary skill in the art.

FIG. 1 is a perspective view illustrating an exemplary staple 100 in accordance with the present disclosure. Although the various parts of exemplary staple 100 are depicted in relative proportion to other parts of staple 100, other configurations in size and orientation of the various parts are also contemplated in other examples. A number of reference directions are illustrated using arrows in FIG. 1 to assist in understanding the details of staple 100. The illustrated directions include: proximal direction P, distal direction D, first laterally outward direction LOA, second laterally outward direction LOB, first laterally inward direction LIA, and second laterally inward direction LIB.

In some examples, staple 100 comprises first arm 102A, second arm 102B, and bridge 104. Bridge 104 may abut, or extend from or adjacent to, the proximal end of first arm 102A to the proximal end of second arm 102B. First arm 102A may include first trunk 106A, with first trunk 106A generally having a greater width than the rest of first arm 102A as depicted in FIG. 1. In some examples, first arm 102A may also include non-trunk portion 105A. The length of first trunk 106A relative to the overall length of first arm 102A can vary in different examples. For instance, first trunk 106A can extend for the entire length of first arm 102A such that bridge 104 abuts with or is adjacent to first trunk 106A. In other examples, first arm 102A may not include first trunk 106A. That is, first arm 102A may not have a portion with

a greater width than the rest of first arm 102A. In such examples, first arm 102A may still have non-trunk portion 105A.

Similarly, second arm 102B may include second trunk 106B, with second trunk 106B generally having a greater width than the rest of second arm 102B. Additionally, second trunk 106B may extend for at least a portion of second arm 102B. A distal portion of second arm 102B may abut the proximal end of second trunk 106B and second arm 102B may further include non-trunk portion 105B. As with first trunk 106A, second trunk 106B may extend along second arm 102B for varying lengths. Additionally, in some examples, second arm 102B may not have a portion with a greater width than the rest of second arm 102B. In FIG. 1, first trunk 106A and second trunk 106B are shown extending distally from a proximal portion of first arm 102A and second arm 102B, respectively.

In the example of FIG. 1, first trunk 106A has a lateral extent, or cross sectional area, that is larger than a lateral extent of the non-trunk portion 105A of first arm 102A and bridge 104. Staple 100 may include a first change in lateral stiffness 108A disposed where the distal end of non-trunk portion 105A of first arm 102A abuts first trunk 106A. As depicted, the change in stiffness is abrupt, but can be gradual in alternative examples such as through a gradual change in lateral extent between first trunk 106A and non-trunk portion 105A. In an example where first trunk 106A extends for the full length of the first arm 102A, the change in stiffness may occur where first trunk 106A abuts bridge 104. With reference to the example of FIG. 1, it will be appreciated that first trunk 106A is mounted eccentrically to first arm 102A and second trunk 106B is mounted eccentrically to second arm 102B. As with first trunk 106A, second trunk 106B has a lateral extent, or cross sectional area that is larger than a lateral extent of second arm 102B or bridge 104. Staple 100 may include a second change in lateral stiffness 108B where the distal end of second arm 102B abuts second trunk 106A. Similarly to first arm 102A, in some examples the change in stiffness may be abrupt or gradual. If second trunk 106B extends for the entire length of second arm 102B, the change in stiffness may occur at the abutment with bridge 104. In additional examples where there may be no change in lateral extent between first and second trunks 106A, 106B and first and second arms 102A, 102B, a change in stiffness may be accomplished by the use of different materials for first and second trunks 106A, 106B and first and second arms 102A, 102B.

Some examples of staple 100 may include at least a first projection 122A, 122C and a second projection 122B, 122D, on each of first trunk 106A and second trunk 106B, respectively. First projection 122A, 122C on each trunk 106A, 106B may further include first proximal surface 124A, 124C extending away from each trunk in a first direction, such as out and away from each opposite trunk 106A, 106B. The first direction may be a direction such that first proximal surface 124A, 124C will engage with tissue or bone after the trunk is inserted therein and by natural movement of the tissue or bone. In some examples, a pullout force may be applied to bridge 104 to further engage first proximal surface 124A, 124C with bone or tissue. The natural movement of the bone or tissue or the pullout force creates a first moment centered on the area of reduced stiffness adjacent each trunk, tending to rotate each trunk thereabout. The rotation of each trunk may further provide a greater holding force of staple 100 in bone or tissue. Second projection 122B, 122D may include second proximal surface 124B, 124D extending away from each trunk in a second direction, different from

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the first direction, such as inward, toward the opposite trunk. For example, the second direction may be selected such that second proximal surfaces **124B**, **124D** will engage tissue or bone after each trunk is inserted therein and by natural movement of the tissue or bone. In some examples, a pullout force may be applied to bridge **104**. A slit or area of reduced cross section in the trunk adjacent the second projections provide an area of weakness so that a second moment is applied to the trunk in response to natural movement of the tissue or bone and/or to a pullout force on bridge **104**. This moment causes rotation of the trunk about the area of weakness and increases the holding force of staple **100**.

As illustrated in the example of staple **100** in FIG. **1**, first trunk **106A** includes a first projection **122A** disposed at an outer side of trunk **106A** and a second projection **122B** disposed at an inner side of the trunk. First projection **122A** includes a first proximal surface **124A** extending away from first trunk **106A** in a first direction. With reference to FIG. **1**, it will be appreciated that the first direction has an outward lateral component and a proximal component so that first proximal surface **124A** extends outwardly and proximally away from first trunk **106A**. For example, the first direction may be selected such that first proximal surface **124A** will engage tissue or bone proximate the outer side of first trunk **106A** after being inserted therein so that a first moment is applied to the trunk in response to natural movement of the tissue or bone and/or to a pullout force on bridge **104**. The moment centers on the arm portion of lesser stiffness adjacent the first projection.

In the example of FIG. **1**, first trunk **106A** includes a first localized area of weakness **120A** disposed proximate second projection **122B**. Second projection **122B** includes a second proximal surface **124B** extending away from first trunk **106A** in a second direction. The second direction is selected such that second proximal surface **124A** will engage tissue or bone proximate the inner side of first trunk **106A** when inserted therein so that a second moment is applied to the trunk in response to natural movement of the tissue or bone and/or a pullout force on bridge **104**. The moment centers around the area of weakness **120A**. The second moment has a direction that is generally opposite a direction of the first moment. It will be appreciated that the second direction has an inward lateral component and a proximal component so that second proximal surface **124B** extends inwardly and proximally away from first trunk **106A**. In other examples, first arm **102A** may not include second projection **122B**. In such examples, only a first moment may be applied to first trunk **106A** in response to natural movement of the tissue or bone and/or a pullout force on bridge **104**.

Second trunk **106B** includes a third projection **122C** disposed at an outer side of second trunk **106B** and a fourth projection **122D** disposed at an inner side of the trunk. In the example of FIG. **1**, third projection **122C** includes a third proximal surface **124C** extending away from second trunk **106B** in a third direction. With reference to FIG. **1**, it will be appreciated that the third direction has an outward lateral component and a proximal component so that third proximal surface **124C** extends outwardly and proximally away from second trunk **106B**. The third direction is selected such that third proximal surface **124C** will engage tissue or bone proximate the outer side of second trunk **106B** when inserted therein so that a third moment is applied to the trunk in response to natural movement of the tissue or bone and/or a pullout force on bridge **104**.

In the example of FIG. **1**, second trunk **106B** includes a second localized area of weakness **120B** disposed proximate fourth projection **122D**. Fourth projection **122D** includes a

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fourth proximal surface **124D** extending away from second trunk **106B** in a fourth direction. In the example of FIG. **1**, the fourth direction is selected such that second proximal surface **124A** will engage tissue or bone proximate the inner side of second trunk **106B** when inserted therein so that a fourth moment is applied to the trunk in response to natural movement of the tissue or bone and/or a pullout force on bridge **104**. The fourth moment has a direction that is generally opposite a direction of the third moment. It will be appreciated that the fourth direction has an inward lateral component and a proximal component so that fourth proximal surface **124D** extends inwardly and proximally away from second trunk **106B**. In other examples, second arm **102B** may not include second projection **122D**. In such examples, only a first moment may be applied to second trunk **106B** in response to natural movement of the tissue or bone and/or a pullout force on bridge **104**.

As depicted in FIG. **1**, the staple **100** includes proximal projections that extend away from or outward from the bridge **104**, while the distal projections extend inward or toward the center of the bridge **104**. This creates generally opposing forces in response to tension on the bridge which, in combination with areas of weakness or reduced lateral extent, substantially increases the holding force of the staple in bone as the different portions of the trunks tend to rotate in opposite directions and apply force to an opposing wall in the hole in bone in which the staple is positioned. It is however, understood that other configurations of the projections are possible. In some examples, only two projections are included and they extend in different directions to cause different force responses as tension is applied to the bridge. Additional examples may include varying numbers of projections which produce one or more moments in each of arms **102A**, **102B**.

In some examples, each projection of staple **100** may be clefted to form a plurality of points for greater retention in tissue. In the example of FIG. **1**, first projection **122A** of first trunk **106A** defines a first notch **126A** that divides first projection **122A** into a first sub-projection and a second sub-projection. Second projection **122B** of second trunk **106B** defines a second notch **126B**. In the example of FIG. **1**, second notch **126B** divides second projection **122B** into a first sub-projection and a second sub-projection. Third projection **122C** of second trunk **106B** defines a third notch **126C** that divides third projection **122C** into a first sub-projection and a second sub-projection. Fourth projection **122D** of second trunk **106B** defines a fourth notch **126D** that divides fourth projection **122D** into a first sub-projection and a second sub-projection.

With continued reference to FIG. **1** and further reference to FIGS. **2** and **3**, first trunk **106A** defines a first cavity **128A** and second trunk **106B** defines a second cavity **128B**. In the examples of FIGS. **1**, **2** and **3**, first cavity **128A** extends into first trunk **106A** and second cavity **128B** extends into second trunk **106B**. The cavity is sized to cooperate with a staple delivery device for holding and inserting the staple into tissue or bone, as later described in detail herein. In summary, the staple delivery device includes longitudinally extending stakes that fit within the cavities **128A**, **128B** to hold the staple **100** and push it into position in the tissue as the stake abuts a portion of its respective trunk. In some examples the cavity may extend through a portion of the length of each trunk, as best depicted in FIG. **2** which indicates the distal end of the staple **100** is closed. Alternatively, first cavity **128A** and second cavity **128B** may extend through the entire length of each trunk **106A**, **106B** or other portions of staple **100** in some examples. As illustrated by

the top view of the staple **100** in FIG. **3**, first cavity **128A** and second cavity **128B** each have a generally rectangular or square cross-sectional shape to cooperate with a similarly shaped cross section on a staple delivery device. However, that first cavity **128A** and second cavity **128B** may have various cross-sectional shapes to cooperate with alternative staple delivery device designs without deviating from the spirit and scope of the present disclosure.

FIG. **4** is an alternative perspective view of example staple **100** depicted in FIG. **1**. In particular, FIG. **4** illustrates in phantom the flexing and bending of the trunks **106A** and **106B** after implant in response to natural movement of the tissue or bone and/or to tension applied to the bridge.

The combination of projections, areas of weakness and changes in lateral extent described with respect to FIGS. **1**, **2**, and **3** provide desired flexing, bending and rotating of the trunk in response to natural movement of the tissue or bone and/or to pull out forces on bridge **104**. Together the various components of staple **100** act as tissue retention members. An exemplary deflected shape is shown with dashed lines in FIG. **4**. Forces applied to staple **100** in response to natural movement of the tissue or bone and/or pullout forces applied to bridge **104** may urge staple **100** to assume the deflected shape shown in FIG. **4**. In some additional examples, distally directed forces may be applied on staple **100** using, for example, the staple delivery system shown later and described herein. In some applications, the staple delivery tool may be used to urge first projection **122A** and third projection **122C** into orientations which lock staple **100** into a target tissue. For example, first projection **122A** and third projection **122C** may be rotated so that these projections engage the target tissue. When this is the case, tension extending through bridge **104** of staple **100** may keep first projection **122A** and third projection **122C** in the rotated position. Also when this is the case, the projections may inhibit staple pullout. Further, rotation of any projection causes a rotational force and imparts, within limits defined by the hole in the bone, some rotation to an adjacent portion of the trunk which contacts or engages the wall of the hole in the bone. Increased pullout forces, such as by natural movement of the tissue or bone and/or pullout forces applied to bridge **104**, may result in increasing holding force with this design.

Next referring to FIG. **5**, an exemplary use or application of the staples of the present disclosure is described. FIG. **5** is a stylized anterior view of a patient **20**. For purposes of illustration, a shoulder **22** of patient **20** is shown in cross-section in FIG. **5**. Shoulder **22** includes a humerus **14** and a scapula **12**. In FIG. **5**, a head **24** of humerus **14** can be seen mating with a glenoid fossa of scapula **12** at a glenohumeral joint. With reference to FIG. **5**, it will be appreciated that the glenoid fossa comprises a shallow depression in scapula **12**. The movement of humerus **14** relative to scapula **12** is controlled by a number of muscles including: the deltoid, the supraspinatus, the infraspinatus, the subscapularis, and the teres minor. For purposes of illustration, only the supraspinatus **26** is shown in FIG. **5**.

With reference to FIG. **5**, a distal tendon **28** of the supraspinatus **26** meets humerus **14** at an insertion point. Scapula **12** of shoulder **22** includes an acromium **32**. In FIG. **5**, a subacromial bursa **34** is shown extending between acromium **32** of scapula **12** and head **24** of humerus **14**. Subacromial bursa **34** is shown overlaying supraspinatus **26** as well as supraspinatus tendon **28** and a portion of humerus **14**. Subacromial bursa **34** is one of the hundreds of bursae found the human body. Each bursa comprises a fluid filled

sac. The presence of these bursae in the body reduces friction between bodily tissues.

The exemplary staples or fasteners described herein may be used to affix tendon repair implants to various target tissues. The shoulder depicted in FIG. **5** is one example where a tendon repair implant may be affixed to one or more bones associated with an articulating joint, such as the glenohumeral joint. Additionally, the tendon repair implant may be affixed to one or more tendons to be treated. The tendons to be treated may be torn, partially torn, have internal micro-tears, be untorn, and/or be thinned due to age, injury or overuse. The methods and apparatus of the present disclosure and related devices may provide beneficial therapeutic effect on a patient experiencing joint pain believed to be caused by partial thickness tears and/or internal micro-tears. By applying a tendon-repair implant early before a full tear or other injury develops, the implant may cause the tendon to thicken and/or at least partially repair itself, thereby avoiding more extensive joint damage, pain, and the need for more extensive joint repair surgery.

FIG. **6** is a stylized anterior view of a shoulder **22** including a humerus **14** and a scapula **12**. In FIG. **6**, a head **24** of humerus **14** is shown mating with a glenoid fossa of scapula **12** at a glenohumeral joint. A supraspinatus **26** is also shown in FIG. **6**. This muscle, along with others, controls the movement of humerus **14** relative to scapula **12**. A distal tendon **28** of supraspinatus **26** meets humerus **14** at an insertion point **30**.

As depicted in FIG. **6**, distal tendon **28** includes a first damaged portion **36**. A number of loose tendon fibers **40** in first damaged portion **36** are visible in FIG. **6**. First damaged portion **36** includes a first tear **42** extending partially through distal tendon **28**. First tear **42** may therefore be referred to as a partial thickness tear. With reference to FIG. **6**, first tear **42** begins on the side of distal tendon **28** facing the subacromial bursa (shown in the previous Figure) and ends midway through distal tendon **28**. Accordingly, first tear **42** may be referred to as a bursal side tear.

With reference to FIG. **6**, distal tendon **28** includes a second damaged portion **38** located near insertion point **30**. As illustrated, second damaged portion **38** of distal tendon **28** has become frayed and a number of loose tendon fibers **40** are visible. Second damaged portion **38** of distal tendon **28** includes second tear **44**. Second tear **44** begins on the side of distal tendon **28** facing the center of the humeral head **24**. Accordingly, second damaged portion **38** may be referred to as an articular side tear.

FIG. **6** illustrates sheet-like implant **50** has been placed over the bursal side of distal tendon **28**. Sheet-like implant **50** is affixed to distal tendon **28** by a plurality of tendon staples **51**. Sheet-like implant **50** is affixed to humerus **14** by one or more bone staples **100** in accordance with designs of staples disclosed herein. Sheet-like implant **50** extends over insertion point **30**, first tear **42** and second tear **44**. Some methods in accordance with this disclosure may include placing a tendon repair implant on the bursal side of a tendon regardless of whether the tears being treated are on the bursal side, articular side or within the tendon. In some cases the exact location and nature of the tears being treated may be unknown. A tendon repair implant may be applied to the bursal side of a tendon to treat shoulder pain that is most likely caused by one or more partial thickness tears in the tendon.

FIG. **7** is a stylized perspective view showing a portion of the body **82** of a human patient **20**. Body **82** includes a shoulder **22**. In the example of FIG. **7**, a plurality of cannulas are positioned to access a treatment site within shoulder **22**.

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In some cases, shoulder **22** may be inflated by pumping a continuous flow of saline through shoulder **22** to create a cavity proximate the treatment site. The cannulas shown in FIG. 7 include a first cannula **80A**, a second cannula **80B** and a third cannula **80C**.

In FIG. 7, a sagittal plane SP and a frontal plane FP are shown intersecting body **82**. Sagittal plane SP and frontal plane FP intersect one another at a medial axis MA of body **82**.

With reference to FIG. 7, sagittal plane SP bisects body **82** into a right side **84** and a left side **86**. Also with reference to FIG. 7, frontal plane FP divides body **82** into an anterior portion **92** and a posterior portion **88**. Sagittal plane SP and a frontal plane FP are generally perpendicular to one another. These planes and portions are used to describe the procedures used in various examples.

First cannula **80A** is accessing a treatment site within shoulder **22** using a lateral approach in which first cannula **80A** pierces the outer surface of right side **84** of body **82**. The term lateral approach could also be used to describe situations in which an instrument pierces the outer surface of left side **86** of body **82**. Second cannula **80B** is accessing a treatment site within shoulder **22** using a posterior approach in which second cannula **80B** pierces the outer surface of posterior portion **88** of body **82**. Third cannula **80C** is accessing a treatment site within shoulder **22** using an anterior approach in which third cannula **80C** pierces the outer surface of anterior portion **92** of body **82**.

FIG. 8 is a stylized perspective view illustrating an exemplary procedure for treating a shoulder **22** of a patient **20**. The procedure illustrated in FIG. 8 may include, for example, fixing tendon repair implants to one or more tendons of shoulder **22**. The tendons treated may be torn, partially torn, have internal micro-tears, be untorn, and/or be thinned due to age, injury or overuse.

Shoulder **22** of FIG. 8 has been inflated to create a cavity therein. A fluid supply **52** is pumping a continuous flow of saline into the cavity. This flow of saline exits the cavity via a fluid drain **54**. A camera **56** provides images from inside the cavity. The images provided by camera **56** may be viewed on a display **58**.

Camera **56** may be used to visually inspect the tendons of shoulder **22** for damage. A tendon repair implant in accordance with this disclosure may be affixed to a bursal surface of the tendon regardless of whether there are visible signs of tendon damage.

An implant delivery system **60** can be seen extending from shoulder **22** in FIG. 8. Implant delivery system **60** is extending through a first cannula **80A**. In certain examples, first cannula **80A** can access a treatment site within shoulder **22** using a lateral approach in which first cannula **80A** pierces the outer surface of a right side of the patient's body. In some cases a physician may choose not to use a cannula in conjunction with implant delivery system **60**. In such examples, the implant delivery system may be advanced through tissue. Implant delivery system **60** comprises a sheath that is affixed to a handle. The sheath defines a lumen and a distal opening fluidly communicating with the lumen. In the example of FIG. 8, the distal opening of the sheath has been placed in fluid communication with the cavity created in shoulder **22**.

A tendon repair implant is at least partially disposed in the lumen defined by the sheath of implant delivery system **60**. Implant delivery system **60** can be used to place the tendon repair implant inside shoulder **22**. In some examples, the tendon repair implant is folded into a compact configuration when inside the lumen of the sheath. When this is the case,

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implant delivery system **60** may be used to unfold the tendon repair implant into an expanded shape. Additionally, implant delivery system **60** can be used to hold the tendon repair implant against the tendon.

The tendon repair implant may be affixed to the tendon while it is held against the tendon by implant delivery system **60**. Various attachment elements may be used to fix the tendon-repair implant to the tendon. Examples of attachment elements that may be suitable in some applications include sutures, tissue anchors, bone anchors, and staples. In the example of FIG. 8, the shaft of a fixation tool **70** is shown extending into shoulder **22**. In some examples, fixation tool **70** is capable of fixing the tendon repair implant to the tendon and bone with one or more staples of the present disclosure while the tendon repair implant may be held against the tendon by implant delivery system **60**.

FIG. 9 is a stylized perspective view of a shoulder **22** including a supraspinatus **26** having a distal tendon **28**. With reference to FIG. 9, a tendon repair implant **50** has been affixed to a surface of distal tendon **28**. Tendon repair implant **50** may comprise, for example, various sheet-like structures without deviating from the spirit and scope of the present detailed description. In some examples, the sheet-like structure may comprise a plurality of fibers. The fibers may be interlinked with one another. When this is the case, the sheet-like structure may comprise a plurality of apertures comprising the interstitial spaces between fibers. Various processes may be used to interlink the fibers with one another. Examples of processes that may be suitable in some applications including weaving, knitting, and braiding. In some examples, the sheet-like structure may comprise a laminate including multiple layers of film with each layer of film defining a plurality of micro-machined or formed holes. The sheet-like structure of the tendon repair implant may also comprise a reconstituted collagen material having a porous structure. Additionally, the sheet-like structure of the tendon repair implant may also comprise a plurality of electro-spun nanofiber filaments forming a composite sheet. Additionally, the sheet-like structure may comprise a synthetic sponge material that defines a plurality of pores. The sheet-like structure may also comprise a reticulated foam material. Reticulated foam materials that may be suitable in some applications are available from Biomerix Corporation of Fremont, California which identifies these materials using the trademark BIOMATERIAL™. The sheet-like structure may be circular, oval, oblong, square, rectangular, or other shape configured to suit the target anatomy.

In the examples of FIG. 9, sheet-like implant **50** is affixed to distal tendon **28** by a plurality of tendon staples **51**. Sheet-like implant **50** is affixed to humerus **14** by a plurality of bone staples **100** as described with respect to the exemplary embodiment of FIG. 1 and detailed throughout this disclosure. However, in other examples, as described previously, various attachment elements may be used to fix tendon repair implant **50** to distal tendon **28** without deviating from the spirit and scope of this detailed description. Example attachment elements that may be suitable in some applications include sutures, tissue anchors, bone anchors, and staples.

In some exemplary methods, a plurality of staples may be applied using a fixation tool. After the staples are applied, the fixation tool may be withdrawn from the body of the patient. Distal tendon **28** meets humerus **14** at an insertion point **30**. With reference to FIG. 9, it will be appreciated that sheet-like implant **50** extends over insertion point **30**. Tendon repair implant may be applied to distal tendon **28**, for example, using the procedure illustrated in the previous

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figures. In various examples, staples may straddle the perimeter edge of the sheet-like implant (as shown in FIG. 9), may be applied adjacent to the perimeter, and/or be applied to a central region of the implant. In some examples, the staples may be used to attach the implant to soft tissue and/or to bone.

Staples or fasteners 100, as exemplified in FIG. 1 and described and illustrated herein can be used to attach tissue and implants to bone. In at least some embodiments, the staple is generally flexible and includes areas of relative lateral weakness on the trunks and can further include an increase in flexibility at the transition from the trunk to the non-trunk portion of the arm or the transition from the trunk to the bridge. As described above, these areas of increased flexibility provide improved staple retention as these portions allow flexing and bending in response to increasing pullout forces. With this flexibility, the fasteners cannot be pounded or driven into bone or other tissue as a conventional hard staple would be driven into paper, wood, tissue or bone. Therefore, for application of the staple of the present disclosure to affix tissue or implants to bone, the staple may be included in a kit that may also include a staple delivery device 200 and various inserts, including pilot hole forming insert assembly 270 as disclosed herein.

FIG. 10 is a schematic illustration depicting a perspective views of staple delivery device 200. As shown on several figures used herein to describe various features of staple delivery device 200, the proximal direction is represented by arrow "P" and the distal direction is represented by arrow "D." Further, as shown in FIG. 10, staple delivery device 200 may include handle assembly 201, retraction assembly 230 and sheath assembly 232.

Handle assembly 201 may include a housing 207 and trigger handle 205. In some example, trigger handle 205 may be utilized to apply a removal force to one or more inserts utilized by staple delivery device 200. For example, in some examples trigger handle 205 may rotate and/or pivot relative to housing 207 such that a leveraged force is applied to an insert placed within staple delivery device 200. Additional examples of the trigger handle 205 and the removal forces generated therewith are described in U.S. patent application Ser. No. 14/931,423 filed on Nov. 3, 2015, the disclosure of which is incorporated herein by reference.

Sheath assembly 232 may include sheath 203 (shown in FIG. 11) and one or more position retention members 204 (shown in FIG. 11). The one or more position retention members 204 may be attached to and extend distally from a distal portion of sheath 203. Sheath member 203 may be a hollow tube and be configured to receive one or more inserts into its lumen. For example, sheath member 203 may be configured to receive a pilot hole insert assembly 270.

A proximal portion of sheath assembly 232 may be attached to handle assembly 201. For example, a proximal portion of sheath assembly 232 may extend into a portion of handle assembly 201 and be rigidly fixed within handle assembly 201. In some examples, a proximal portion of the sheath 232 may engage (e.g., mate) with the interior structural elements of handle assembly 201. For example, handle assembly 201 may include a recess or cavity designed to engage and/or mate with the proximal portion of sheath assembly 232. It can be appreciated that in some examples, sheath assembly 232 is fixed relative to handle assembly 201. In other words, in some examples sheath 232 does not move, slide, translate, etc. with respect to handle assembly 201.

Retraction assembly 230 may include cover 202 and retractor member 210. Cover 202 may include a proximal

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portion 211 and a distal portion 212. Further, cover 202 may include a lumen extending the length of cover 202. For example, the lumen extending within cover 202 may extend along the longitudinal axis of cover 202 from proximal portion 211 to distal portion 212.

The distal portion 212 of cover 202 may include one or more leaflets 215. For example, the distal portion 212 of cover 202 may include 1, 2, 3, 4, 5, 6, 7, 8, 9, 10 or more leaflets. In some instances (such as that shown in FIG. 10), leaflets 215 may be positioned on the distal end of cover 202.

In some instances, the proximal portion 211 of cover 202 may be attached to a distal portion of retractor member 210. For example, FIG. 10 shows proximal portion 211 of cover 202 extending into distal portion 223 of retractor member 210. In some examples, a proximal portion of the cover 202 may engage (e.g., mate) with the interior structural elements of retractor member 210. For example, retractor member 210 may include a recess or cavity designed to engage and/or mate with the proximal portion of retractor member 210. It can be appreciated that in some examples, cover 202 is fixed relative to retractor member 210. In other words, in some examples cover 202 and retractor member 210 move, slide, and/or translate as a combined assembly. Additionally, in some instances cover 202 may extend over a portion of sheath member 203. In other words, in some examples, sheath 203 may be positioned within the lumen of cover 202.

As illustrated by FIGS. 10 and 11, in some examples retraction assembly 230 may slide, move, translate and/or shift with respect to sheath assembly 232 and handle assembly 201. Specifically, staple delivery device 200 may include a first configuration (e.g., position) in which the cover 202 of retraction assembly 232 encloses (e.g., covers) sheath assembly 232. This configuration may correspond to the retraction assembly 230 being moveable along handle assembly 201 to a distal position along handle assembly 201. Further, staple delivery device 200 may include a second configuration in which the cover 202 of retraction assembly 230 does not completely enclose sheath assembly 232. As shown in FIG. 11, the second configuration may correspond to retraction assembly 230 being moveable proximally along handle assembly 201. In other words, the retraction assembly 230 (e.g., retractor member 210 and cover 202) may slide, move or shift as a single piece along handle assembly 201. Further, when retraction assembly 230 moves from a distal position to a proximal position, the distal portion 212 of cover 202 may be positioned proximal of a distal portion of sheath member 203 (thereby "uncovering" a distal portion of sheath 203).

As shown in FIG. 11, leaflets 215 may hinge, flex, pivot, rotate and/or expand outward in a radial direction, thereby creating an opening through which sheath assembly 232 may extend. For example, as retraction assembly 230 moves from a first, distal position to a second, proximal position, leaflets 215 may flare open (as shown in FIG. 11) as the distal portion 212 of cover 202 moves over the distal portion of the sheath assembly 232.

Additionally, leaflets 215 may include a proximal portion 217 and a distal portion 219. In some examples, one or more of leaflets 215 may taper from proximal portion 217 to distal portion 219. Further, each leaflet 215 may curve inward toward longitudinal axis 221 (shown in FIG. 12) and taper as each leaflet 215 extends distally. In at least some embodiments, cover 202 comprises an even number of leaflets 215. More specifically, in at least some embodiments, cover 202 comprises four, six, eight, ten, or any other suitable number of leaflets 215. However, in other embodiments, cover 202 may comprise an odd number of leaflets 215.

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In examples where cover **202** comprises an even number of leaflets **215**, as staple delivery system **200** is being inserted into an incision, opposite leaflets **215** may collapse against each other when being advanced through the incision and into tissue. This may help prevent tissue from entering cover **202**. For instance, each leaflet **215** may have a thickness extending from an outer surface to an inner surface. The leaflet thickness may vary in different embodiments between about 0.05 inches (1.27 mm) and about 0.15 inches (3.81 mm). As a force is applied to the outer surface of leaflets **215** and leaflets **215** collapse together such that one leaflet **215** converges with an adjacent face of an adjacent leaflet **215**, leaflets **215** may become pressed together. In this manner, each of leaflets **215** may support each other when a force is applied to the outer surface of leaflets **215**. The thickness and configuration of leaflets **215** may help to prevent leaflets **215** from buckling inward when forces are applied to the outside surface of leaflets **215**. With leaflets **215** pressed together (e.g. abutting one another), leaflets **215** may form a solid plug which prevents tissue from entering cover **202** as staple delivery system **200** is advanced through tissue. Additionally, when leaflets **215** are pressed together under an external force, leaflets **215** may translate such forces into a force acting in a substantially axial direction (e.g., along central longitudinal axis **221**).

As discussed above, when a force is applied to the outer surface of leaflets **215**, leaflets **215** are configured to collapse together. However, when a force is applied to an inner surface of leaflets **215** (e.g., from an interior of the cover **202**), the force may push leaflets **215** in an outward direction away from each other. If the force is large enough, leaflets **215** flare radially outward and diverge from one another, exposing sheath assembly **232**.

Additionally, while FIGS. **10** and **11** show leaflets **215** including a cavity or recess formed in the sidewall thereof, it is contemplated that leaflets **215** may include a variety of geometric shapes and/or configurations tailored to a specific functionality.

For example, it may be desirable for a clinician to insert staple delivery device **200** through an access site of a patient while cover **202** is positioned over sheath assembly **232**. Specifically, cover **202** (including leaflets **215**) may allow position retention members **204** to slide past tissue (e.g., access site) without getting caught on skin. In other words, cover **202** may protect, or guard against, position retention members **204** from getting caught on the tissue. For example, the tissue defining the access site may slide around the curved surfaces of cover **202** (including leaflets **215**), thereby allowing the clinician to advance staple delivery device **200** through the tissue without any portions of staple delivery device **200** catching on tissue.

The ability of cover **202** to prevent portions of staple delivery device **200** from catching on tissue may be desirable for a variety of reasons. For example, cover **202** may permit a clinician to use the staple delivery device **200** independent of a separate access cannula. As stated above, cover **202** may prevent position retention members **204** from catching on tissue (similar to the functionality provided by a separate access cannula). As the user advances staple delivery device **200** into the patient, the tissue may slide along cover **202** and past position retention members **204**. In this manner, a user may maneuver staple delivery device **200** to an implant site without catching position retention members **204** on tissue of the patient causing unwanted tissue damage.

FIGS. **12** and **13** illustrate alternative views of staple delivery device **200**. Namely, FIGS. **12** and **13** show sheath

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assembly **232** including sheath **203** and position retention members **204** extending within a portion of cover **202** (depicted by dotted line **213**). As shown, sheath assembly **232** includes a longitudinal axis **221**. It can be appreciated that cover **202** and sheath member **203** may be coaxial with one another along the longitudinal axis **221**. Further, FIG. **213** illustrates that retractor member **210** may move relative to handle assembly **201** along a line which is substantially parallel to or aligned with the longitudinal axis **221**. However, this is not intended to be limiting. Other means are contemplated for moving retractor member **210** with respect to handle assembly **201**. For example, it can be appreciated that retractor **210** may rotate, pivot, spin and/or swing away from handle assembly **201** and/or sheath assembly **232**. These alternative methods may retract the retractor member **210** to achieve substantially the same results (e.g., uncovering the distal portion of the sheath assembly **232**) as those discussed above with respect to the linear retraction of the retractor member **210**.

It can be appreciated that in order for retractor member **210** (along with cover **202**) to move relative to handle assembly **201**, one or more forces needs to be imparted to the retractor member **210** such that it moves from a distal position to a proximal position (in which the sheath assembly extends through cover **202** as described above). In some examples, a spring (either in tension or compression) may be utilized to provide the linear force necessary to move the retractor member **210** relative to handle assembly **201**.

FIG. **14** shows staple delivery device **200** including spring **240** positioned within handle assembly **201**. While spring **240** is shown in FIG. **14** to be generally in tension, it is contemplated that alternative embodiments could incorporate spring **240** in compression.

It can be appreciated from FIG. **14** that spring **240** may bias retractor member **210** in a proximal position. In other words, spring **240** may “pull” retractor member **210** in a proximal direction to a position in which the sheath assembly **232** extends through cover **202** as described above.

Therefore, it can be appreciated that a locking force may be necessary to maintain the retraction assembly **230** in a distal position (e.g., a position in which cover **202** may enclose sheath assembly **232**). FIG. **14** also shows retraction assembly **230** held in a distal position via actuation member **216**. As shown in FIG. **14**, actuation member **216** may include a projection **222** extending away from actuation member **216**. Further, retractor member **210** may include a recess **220** designed to engage projection **222**. When positioned as shown in FIG. **14**, actuation member **216** (via projection **222**) substantially prevents the proximal movement of retraction assembly **230**.

Additionally, FIG. **14** illustrates that actuation member **216** may rotate about pivot point **218**. Pivot point **218** may be positioned along housing assembly **201**. However, it can be appreciated that pivot point **218** may be positioned such that projection **222** may engage with recess **220**.

FIG. **15** shows actuation member **216** rotated such that projection **222** disengages with recess **220**. As discussed above, disengaging projection **222** from recess **220** may allow retraction assembly **230** to move from the distal position to a proximal position (in which the sheath assembly extends through cover **202** as described above). The arrow shown in FIG. **15** illustrates the movement of the retractor member **210** after the actuation member **216** is disengaged from recess **220**.

Generally, components of staple delivery device **200** may be made from any rigid material. For example, in different examples, staple delivery device may be made from any of

a variety of different metals or metal alloys. Some example metals include the various alloys of stainless steel. In other examples, staple delivery device 200 may be constructed from plastic. In such examples, the plastic may generally be rigid and resist deformation. Some example plastics include polymers such as Nylon 12, Polyethylene terephthalate (PET), polybutylene terephthalate (PBT), Polyamide 12, Polyether block amide (PEbax) 7233, Pebax 7033, PTFE, Polyaryletherketones (PEEK), Polyphenylene Oxide (PPO), high density polyethylene (HDPE) and the like. In still other examples, staple delivery device may be constructed from Ixef® plastics, which generally include glass fiber reinforcement in addition to one or more polymers. In still other examples, some portions of staple delivery device 200 may be constructed from plastic, such as housing 207 and trigger handle 205, and other portions of staple delivery device 200 may be constructed from metal, such as sheath 203. Of course, in other examples, other combinations of components of staple delivery device 200 may be made from plastic and metal.

FIG. 16 depicts position retention members 204 in a close-up fashion. In some examples, position retention members 204 include curved sides 206. In such examples, curved sides of retention members 204 are configured to mate with curved sides 206 of a variety of example inserts (e.g., pilot hole forming insert 270).

As discussed above, in some examples a clinician may insert staple delivery device 200 through an access site while maintaining sheath assembly 232 (including position retention members 204) enclosed within cover 202. After positioning the cover 202 adjacent the target site, the clinician may rotate the actuation member 216, thereby disengaging projection 222 from recess 220. Furthermore, rotation of actuation member 216 may cause cover 202 to retract in a proximal direction and expose the position retention members 204. At this point, the position retention members 204 may be adjacent the target site.

Once position retention members 204 are adjacent the target site, a user may insert pilot hole forming insert 270 into the lumen of sheath 203. FIG. 17 is a schematic illustration of pilot hole forming insert 270. Generally, insert 270 may have proximal head 271, shaft 273, and distal end 275. Additionally, insert 270 may have one or more pilot hole forming members 272 connected to distal end 275. In different examples, pilot hole forming members 272 may take various different shapes, such as spikes, spears, prongs, or other shapes. Whatever shape pilot hole forming members 272 may take, they may generally have pointed distal ends for piercing tissue or bone.

Insert 270 may include proximal head 271. Proximal head 270 may have connecting fins 277 extending lengthwise down insert 270 toward distal end 275. Connecting fins 277 may additionally have inward facing protrusions 279. As a user inserts insert 270 into sheath 203, inward facing protrusions 279 may slide into grooves 214, securing insert 270 to staple delivery device 200.

A user may then position pilot hole forming members 272 at a location of the implant site adjacent to tissue or bone into which a staple will be deployed. FIG. 18 is a cross section illustration and depicts the implant site when sheath 203 with received insert 270 is positioned near the implant site. In FIG. 18, implant 310 can be seen positioned on top of patient tissue 312. Additionally, sheath 203 is positioned adjacent to implant 310 and patient tissue 312 with pilot hole forming members 272 extending distally of position retention members 204.

Once the user has positioned pilot hole forming members 272 adjacent tissue or bone at the implant site, the user may apply a distally directed force to proximal head 271 in the distal direction. As insert 270 moves in the distal direction, pilot hole forming members 272 positioned at the implant site are driven into the tissue or bone. FIG. 19 illustrates the position of proximal head 271 with respect to housing 207 when insert 270 has been advanced in the proximal direction as far as housing 207 will allow. As depicted, no gap exists between proximal head 271 and housing 207.

FIG. 20 is another cross section illustration and depicts the position of pilot hole forming members 272 at the implant site when insert 270 has been advanced distally as far as housing 207 will allow. Pilot hole forming members 272 can be seen extending through implant 310 and into patient tissue 312.

Another feature that staple delivery device may employ and can be seen in FIGS. 18 and 20 is the progressive disclosure of pilot hole forming members 272. That is, when inward facing protrusions 279 are engaged with grooves 214, pilot hole forming members 272 may extend distally at least partially beyond the distal end of position retention members 204. In some examples, in this initial position of proximal head 271 with respect to housing 207, pilot hole forming members 272 may extend between 0.05 inches (1.27 millimeters) and 0.35 inches (8.89 millimeters) beyond the distal end of position retention members 204, and in at least some examples, pilot hole forming members 272 may extend 0.083 inches (2.10 millimeters) beyond the distal end of position retention members 204. As force is applied to proximal head 271, and as insert 270 progresses distally, pilot hole forming members 272 extend progressively more beyond the distal end of position retention members 204. In FIG. 20, pilot hole forming members 272 are fully extended. In this position, pilot hole forming members 272 may extend between 0.4 inches (10.16 millimeters) and 0.65 inches (16.51 millimeters) beyond the distal end of position retention members 204. One advantage to examples that include this progressive disclosure feature is that having less length of pilot hole forming members 272 extending beyond position retention members 204 in the initial position may help prevent pilot hole forming members 272 from bending as force is applied to proximal head 271. In addition to driving pilot hole forming members 272 into tissue 312, the force applied to proximal head 271 may also drive position retention members 204 into tissue 312, as can be seen in FIG. 20.

Once the user has fully driven pilot hole forming members 272 into tissue 312, the user may remove insert 270. To remove insert 270, a user may squeeze trigger handle 205. Although tissue 312 may be applying squeezing forces to pilot hole forming members 272 which work to retain pilot hole forming members 272 in tissue 312, the force multiplication action of staple delivery device 200, as described previously, may assist the user in removing pilot hole forming members 272 from tissue 312.

FIG. 21 is another cross section illustration that depicts the implant area once insert 270 has been removed from staple delivery device 200. FIG. 21 illustrates that even after insert 270 has been removed, position retention members 204 may still remain in tissue 312. Position retention members 204 may act to maintain sheath 203 in position with respect to pilot holes 309 left by the pilot hole forming members 272.

Next, a user may insert staple delivery insert 600, as depicted in FIG. 22, into staple delivery device 200. Again, staple delivery insert 600 may generally be similar to insert

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270. For instance, insert **600** may be comprised of shaft **602**, proximal head **604**, and distal end **606**. However, staple delivery insert **600** may have arms **608** connected to distal end **606** which may retain a staple, such as staple **100**. In some examples, arms **608** may include detents **610**, as illustrated in FIG. 23. Arms **608** may be designed to be received into cavities **128A**, **128B** of staple **100** and retain staple **100** with friction. For instance, when arms **608** are received within cavities **128A**, **128B** of staple **100**, detents **610** may press against inner surfaces of cavities **128A**, **128B**, thereby retaining staple **100** to arms **608** by friction. Once staple delivery insert **600** is received within sheath **203**, a user may then apply force to the proximal end of staple delivery insert **600**. The applied force may drive arms **610** of the staple delivery device, along with retained staple **100**, into pilot holes **309**. As discussed with respect to FIGS. 1-4, natural movement of tissue **312** and/or a pullout force applied to the bridge of staple **100** may act to secure staple **100** within tissue **312**.

The user may then remove staple delivery insert **600** from staple delivery device **200**. Tissue **312** may impart a holding force on staple **100** sufficient to overcome the friction force between detents **610** on arms **608** of staple delivery insert **600** and staple **100** such that staple delivery insert **600** may be removed from tissue **312** while staple **100** remains in tissue **312**, as depicted in FIG. 24.

Finally, the user may then retract staple delivery device **200** from the patient and finish the procedure to secure implant **310** to tissue **312** of the patient. This may include fixing staple **100** to a tendon of the patient with one or more fixation devices. Alternatively, implant **310** may have already been affixed to the tendon before affixing implant **310** to tissue **312**.

In light of the above description, it should be understood that other examples of staple **100**, staple delivery device **200**, and insert **270** that are still within the spirit and scope of the present disclosure may differ from the specific examples illustrated herein.

Accordingly, it should be generally understood that even though numerous characteristics of various embodiments have been set forth in the foregoing description, together with details of the structure and function of various embodiments, this detailed description is illustrative only, and changes may be made in detail, especially in matters of structure and arrangements of parts illustrated by the various embodiments to the full extent indicated by the broad general meaning of the terms in which the appended claims are expressed.

What is claimed is:

1. A fastener delivery tool, comprising:
 - a sheath assembly including a tubular sheath and one or more position retention members;
 - a retraction assembly including a cover and a retractor member, which move in unison between a first position of the retraction assembly and a second position of the retraction assembly, wherein the cover includes a lumen extending therein;
 - a handle assembly coupled to a proximal end of the sheath assembly, the handle assembly including a housing fixed to a proximal end of the sheath and an actuation member, the actuation member including a projection configured to engage with a recess disposed along the retractor member;
 - wherein the sheath assembly extends through the lumen of the cover; and
 - wherein actuation of the actuation member moves the retractor assembly along the housing of the handle

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assembly from the first position to the second position, wherein the retractor member is configured to move relative to the housing between the first position of the retraction assembly in which the sheath assembly is disposed within a distal portion of the cover and the second position of the retraction assembly in which the sheath assembly extends out of the distal portion of the cover.

2. The fastener delivery tool of claim 1, wherein the actuation member is designed to rotate relative to the housing.

3. The fastener delivery tool of claim 1, wherein actuation of the actuation member rotates the actuation member about an attachment point disposed along the housing, and wherein rotation of the actuation member about the attachment point is designed to disengage the projection from the recess.

4. The fastener delivery tool of claim 1, wherein the one or more position retention members fixedly extend from a distal end of the tubular sheath to maintain a position of the tubular sheath at a target site.

5. The fastener delivery tool of claim 1, wherein the retraction assembly is biased to be in the second position.

6. The fastener delivery tool of claim 1, further comprising a spring coupled to the retractor member.

7. The fastener delivery tool of claim 6, wherein the spring shifts the retractor member from the first position to the second position when the actuation member is actuated.

8. The fastener delivery tool of claim 1, wherein the cover includes one or more leaflets positioned on a distal portion thereof.

9. The fastener delivery tool of claim 8, wherein at least one of the one or more leaflets includes a proximal portion and a distal portion, and wherein at least one of the one or more leaflets are tapered from the proximal portion to the distal portion.

10. The fastener delivery tool of claim 8, wherein the one or more leaflets are biased in a closed position.

11. The fastener delivery tool of claim 8, wherein the one or more leaflets are configured to expand radially outward as the retraction assembly slides from the first position to the second position.

12. The fastener delivery tool of claim 1, wherein shifting the retractor member between the first position and the second position uncovers the sheath assembly in vivo.

13. The fastener delivery tool of claim 1, wherein a proximal portion of the cover is attached to the retractor member.

14. A fastener delivery tool, comprising:

a handle assembly including a housing, a retractor member and an actuation member; and

a sheath assembly including a sheath, a cover and a position retention member;

wherein the sheath extends within a lumen of the cover; wherein a proximal portion of the sheath assembly is coupled to the handle assembly;

wherein the actuation member includes a projection configured to engage with a recess disposed along the retractor member;

wherein actuation of the actuation member slides the retractor member from a first position in which the sheath assembly is disposed within a distal portion of the cover to a second position in which the sheath assembly extends out of the distal portion of the cover.

15. The fastener delivery tool of claim 14, wherein actuation of the actuation member rotates the actuation member about an attachment point disposed along the

housing, and wherein rotation of the actuation member about the attachment point is designed to disengage the projection from the recess.

16. The fastener delivery tool of claim 14, wherein the retractor member is biased to be in the second position. 5

17. The fastener delivery tool of claim 14, wherein the cover includes one or more leaflets positioned on a distal portion of the cover.

18. A method for deploying a fastener, the method comprising: 10

positioning a fastener delivery tool adjacent a target site, the fastener delivery tool comprising:

a sheath assembly including a sheath and a position retention member;

a retraction assembly including a cover and a retractor member, wherein the cover includes a lumen extending therein; and 15

a handle assembly coupled to a proximal end of the sheath assembly, the handle assembly including a housing, a trigger and an actuation member; 20

wherein the sheath assembly extends through the lumen of the cover;

actuating the actuation member to move the retractor member from a first position in which the sheath assembly is disposed within a distal portion of the cover to a second position in which the sheath assembly extends out of the distal portion of the cover, wherein the actuation member includes a projection configured to engage with a recess disposed along the retractor member; and 25 30

deploying a staple along a target site.

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